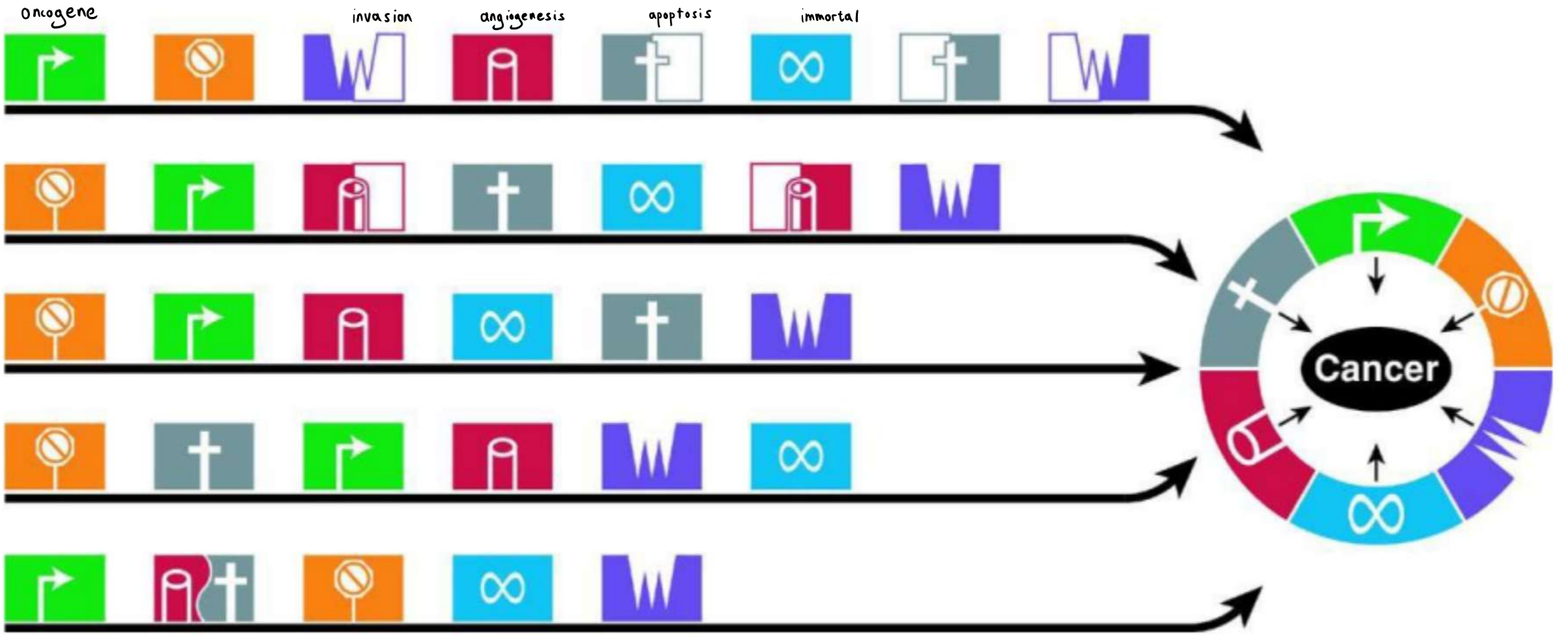
Carcinogenesis

Ashley Soosay

WHAT IS CARCINOGENESIS?

- The noun "carcinoma" and the suffix "-genesis“
- First used in 1923 in The Lancet.
- Carcinoma - A malignant tumor. (Greek word “karkinos”)
- -gen – *suffix* meaning “something produced”

- The word "carcinogenesis" means the production of cancer.


Oncogene
invasion
angiogenesis
apoptosis
immortal
Cancer

Objectives

- List and describe succinctly the *10 Hallmarks of Cancer*.
- *Distinguish a cancer cell* from a normal cell based on hallmarks of cancer.
- *Differentiate the contribution of genes* to the risk and growth of cancer.
- Explain why *cancer cells* are so *efficient* in *survival, proliferation and metastases*.

TERMINOLOGIES

Apoptosis: Programmed cell death

Mitosis: Cell division to create identical daughter cells

Telomeres: Nucleotide sequences at the end of chromosomes that shorten after cell division

Angiogenesis: The creation of new blood vessels

Tumorigenesis: The creation of tumors

Metastasis: Movement of cancer from its origin to another part of the body

TME: Tumor Microenvironment

Exosome: Extracellular vesicles which are precursors to cancer metastasis

Epigenetic

- Epigenetic acetylation - involves adding acetyl groups to histones, which loosens DNA-histone packing, opens chromatin, and **activates gene expression**, contrasting with DNA methylation.
- Catalyzed by **Histone Acetyltransferases (HATs)** and reversed by **Histone Deacetylases (HDACs),** this dynamic process regulates gene access for transcription factors, influencing cell function, development, and diseases like cancer and neurological disorders, making it a key therapeutic target.

Scanned with CamScanner

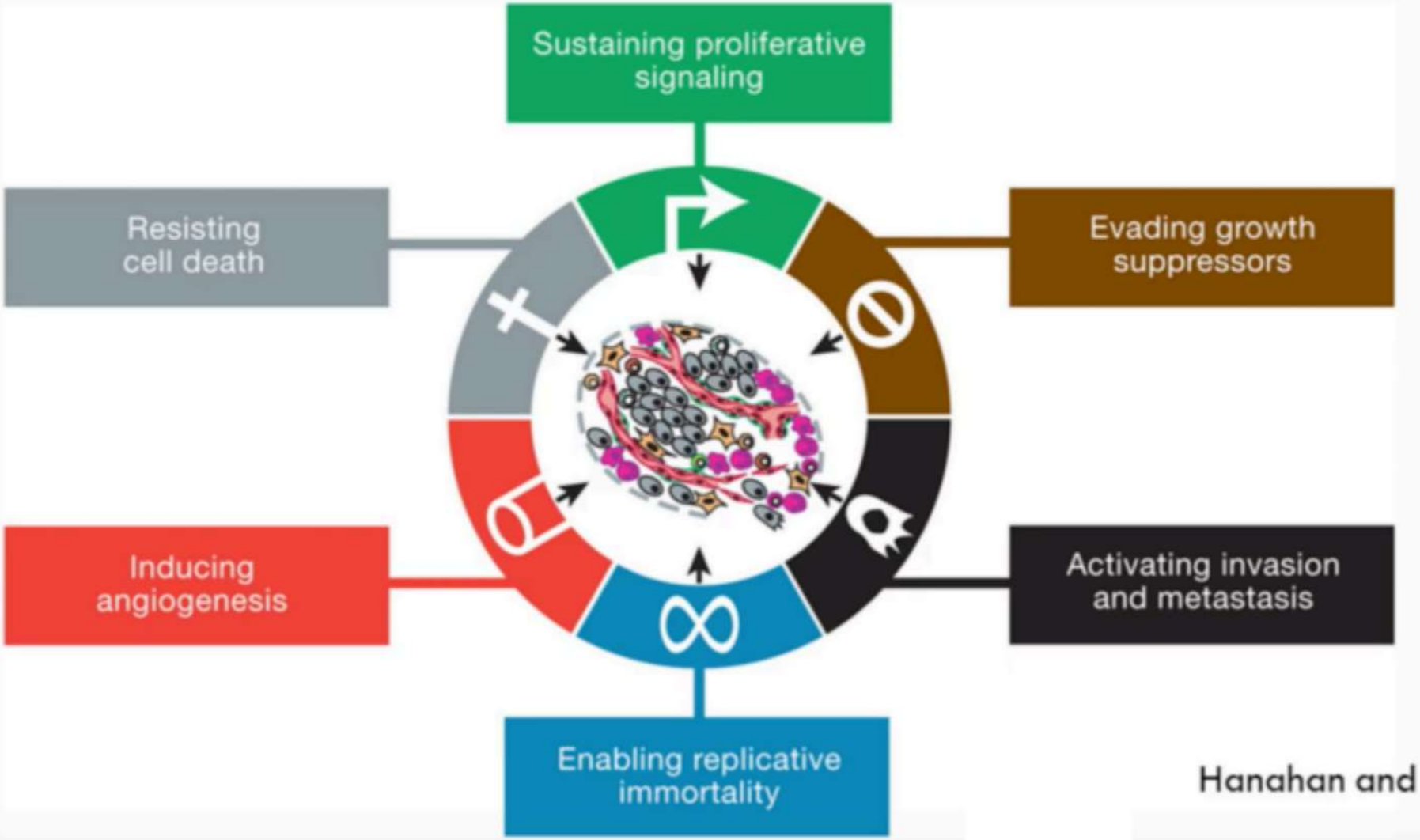
Epigenetic acetylation

- **Mechanism**: Acetyl groups (from acetyl-CoA) attach to positively charged lysine residues on **histones**, neutralizing their charge.
- **Chromatin Structure**: This neutralization causes histones to repel DNA (which is negatively charged), leading to unwound, "open" chromatin (euchromatin).
- **Gene Regulation**: Open chromatin allows transcription machinery (like RNA polymerase) easier access to DNA, turning genes "on" (transcriptional activation).
- **Enzymes**: Histone Acetyltransferases (HATs) add acetyl groups, while Histone Deacetylases (HDACs) remove them, controlling the balance.
- **Biological Significance**: Crucial for normal cell functions, cell cycle, differentiation, learning, memory, and implicated in diseases like Alzheimer's, Parkinson's, and cancer.

Scanned with CamScanner

THE HALLMARKS OF CANCER 1.0

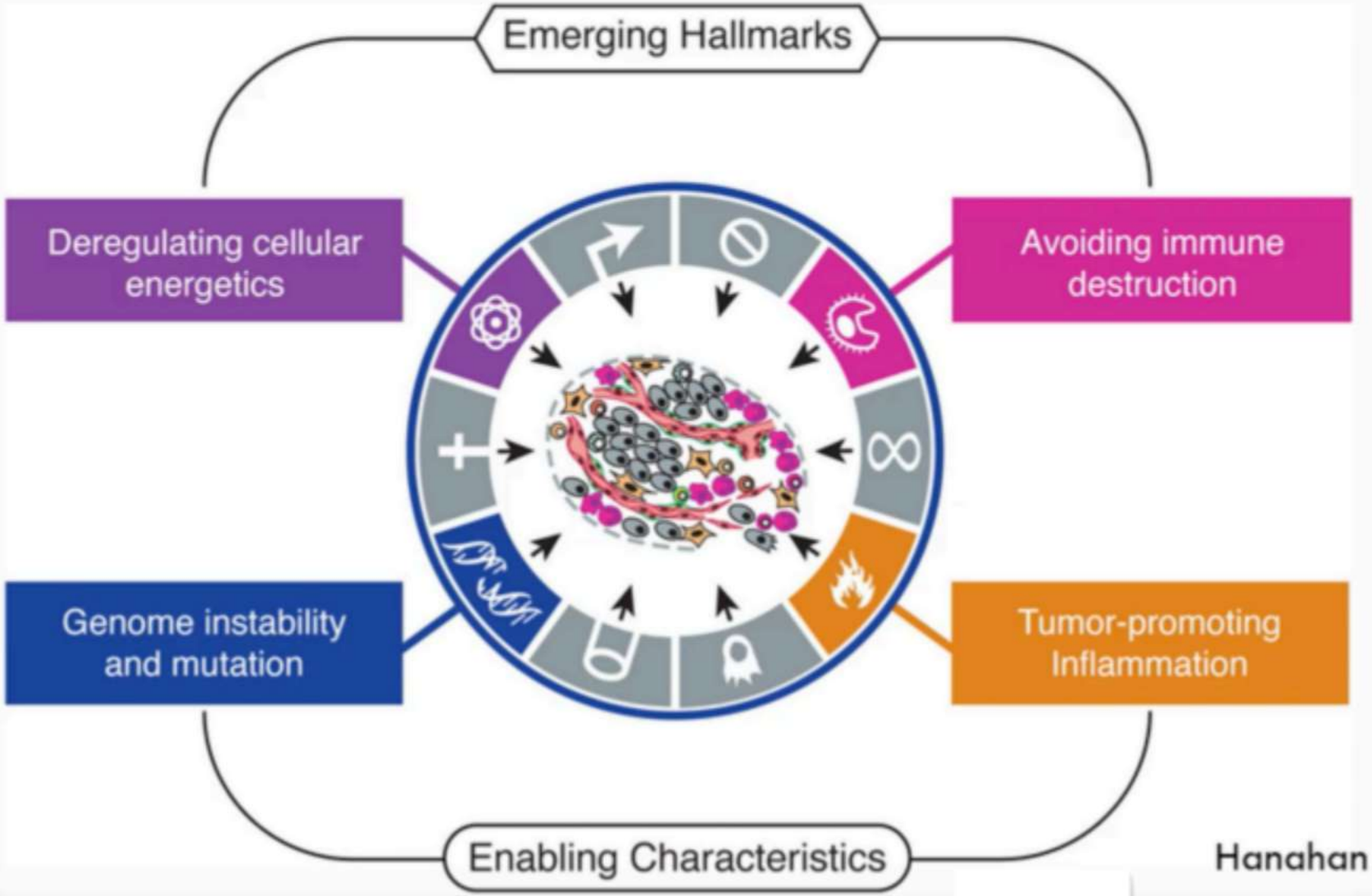
apoptosis → eliminate rogue cell



Hanahan and Weinberg (2000)

THE HALLMARKS OF CANCER 2.0

Warburg effect

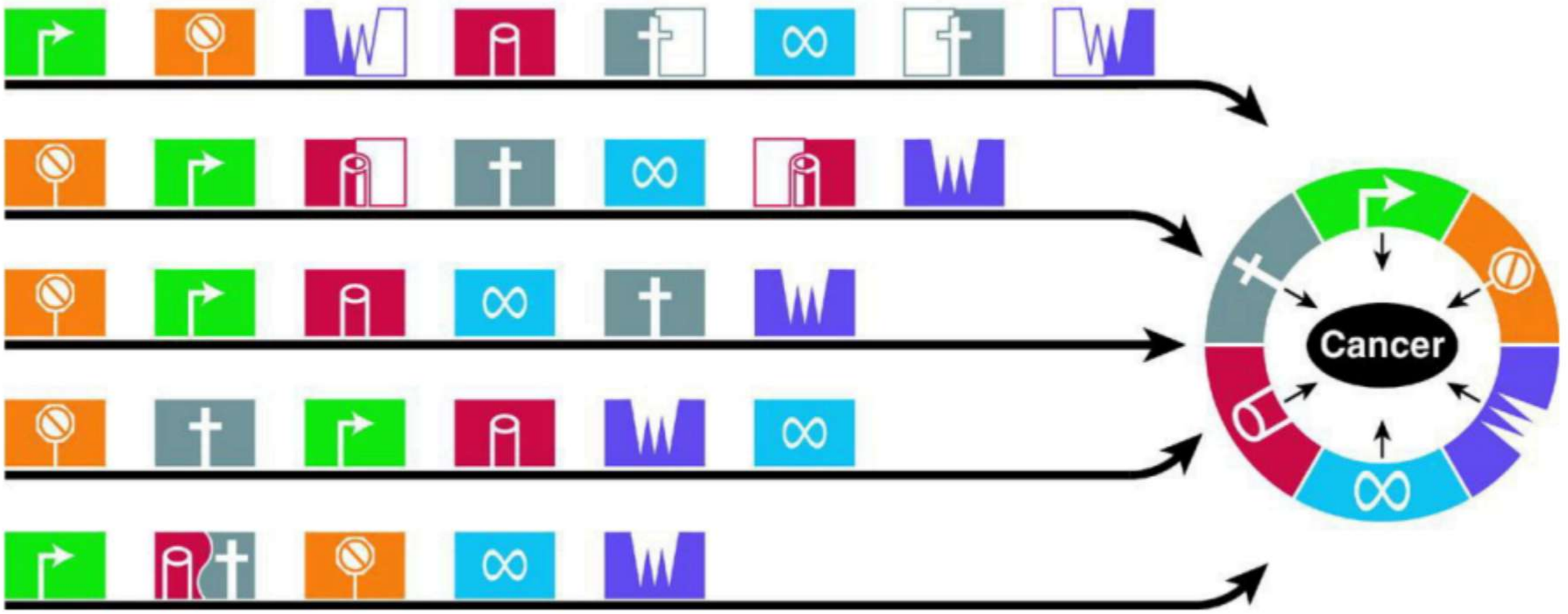


Hanahan and Weinberg (2011)

Hallmarks of cancer

1. Sustaining Proliferative Signaling
2. Evading Growth Suppressors
3. Resisting Cell Death
4. Inducing Angiogenesis
5. Activating Invasion & Metastasis
6. Enabling Replicative Immortality

1. Deregulating Cellular Energetics
2. Avoiding Immune Destruction
3. Genome Instability & Mutation
4. Tumour Promoting Inflammation


Cancer

Component	Acquired Capability	Example of Mechanism	
	Self-sufficiency in growth signals	Activate H-Ras oncogene	
	Insensitivity to anti-growth signals	Lose retinoblastoma suppressor	
	Evading apoptosis (2 pathway: dead cell receptor; mitochondria) — relationship?	Produce IGF survival factors	
	Limitless replicative potential	Turn on telomerase	
	Sustained angiogenesis	Produce VEGF inducer	
	Tissue invasion & metastasis	Inactivate E-cadherin	

Making Sense of Carcinogenesis

HeLa cells origin problem?

1. Enabling Replicative Immortality
2. Genome Instability & Mutation
3. Evading Growth Suppressors
4. Resisting Cell Death
5. Sustaining Proliferative Signaling

1. Deregulating Cellular Energetics
2. Avoiding Immune Destruction
3. Tumour Promoting Inflammation
4. Inducing Angiogenesis
5. Activating Invasion & Metastasis

ENABLING REPLICATIVE IMMORTALITY

HALLMARK ONE

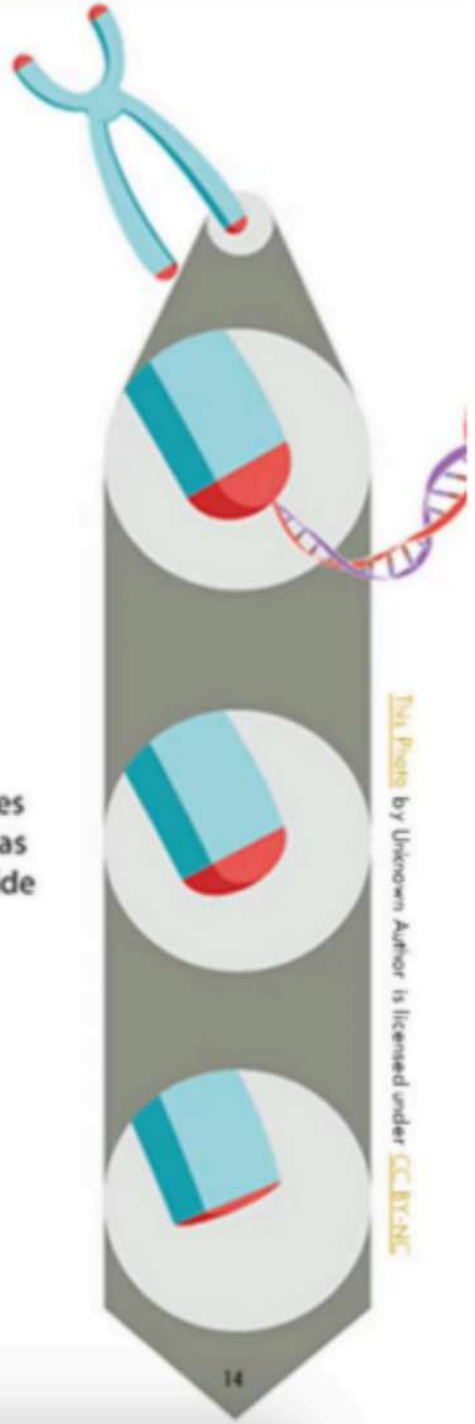
HALLMARK I: REPLICATIVE IMMORTALITY

The End Replication Problem

- Human cells undergo mitosis a finite number of times
- Telomeres shorten after each mitotic division
- Hayflick's limit initiates cell senescence (G0 phase)

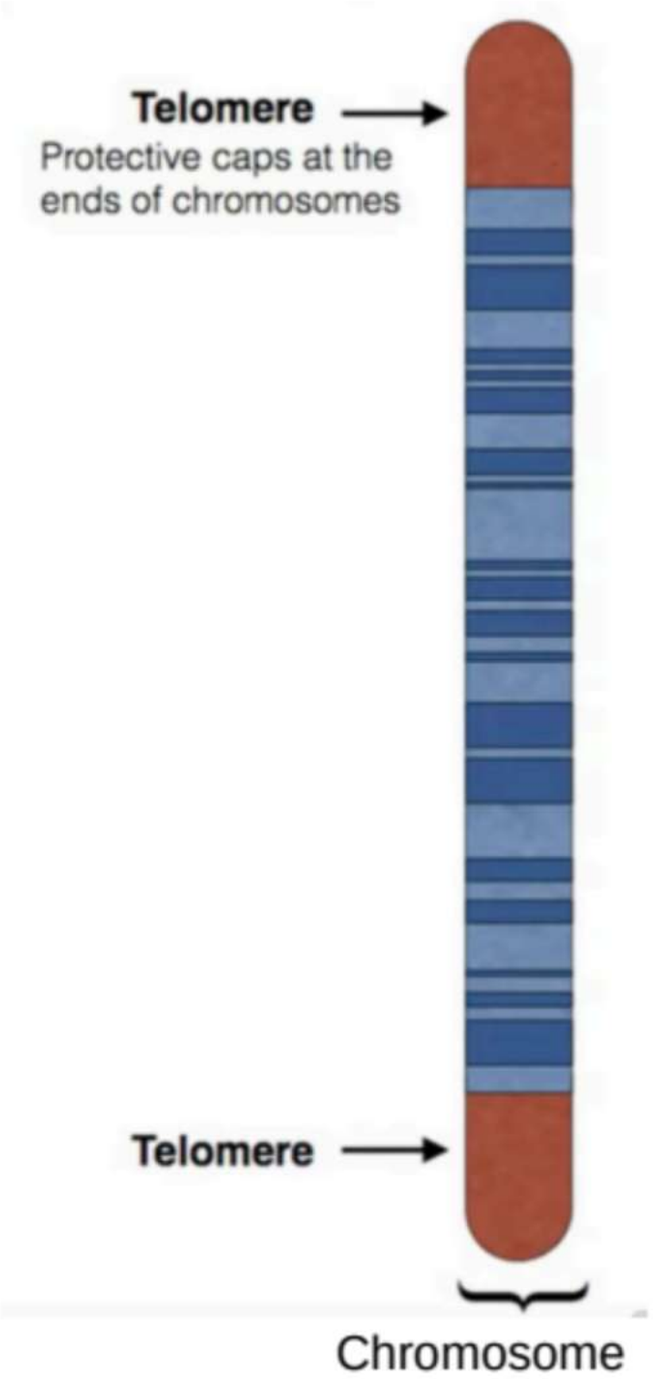
Cancer cells exceed Hayflick's limit by elongating their telomers using Telomerase

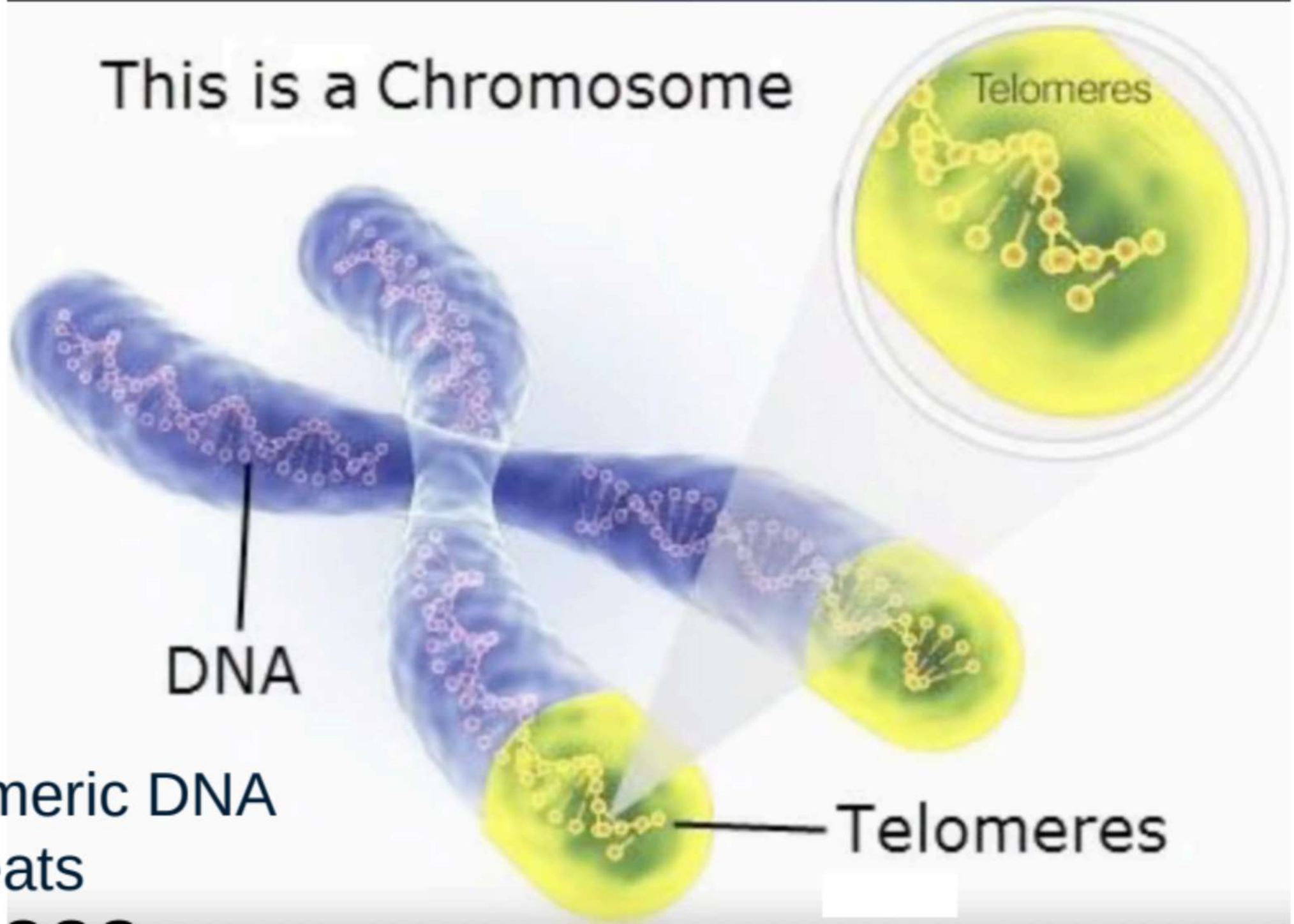
If a cell has an increased expression of Telomerase, it is a sign that the cell is cancerous



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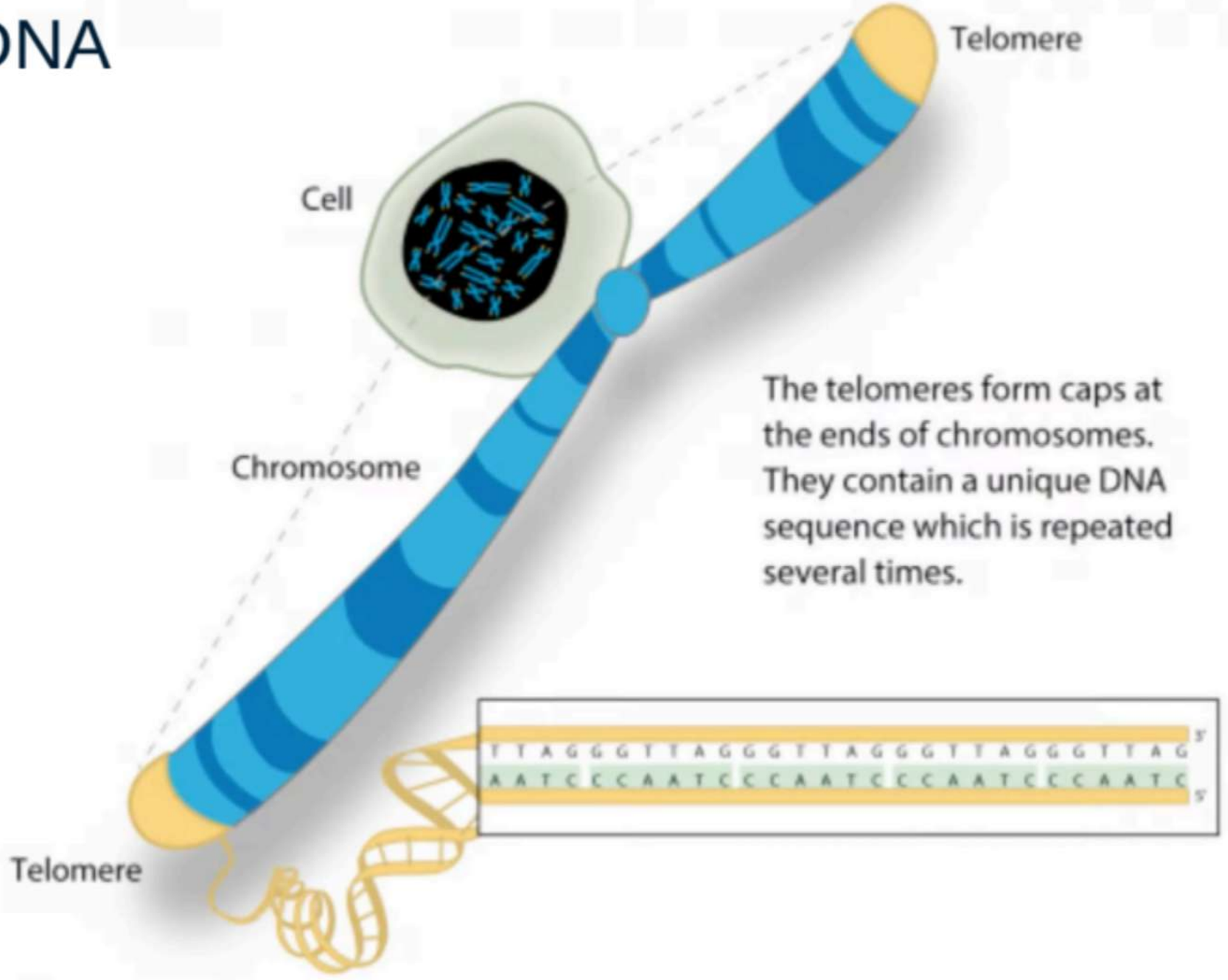
telomeric DNA repeats **TTAGGG**



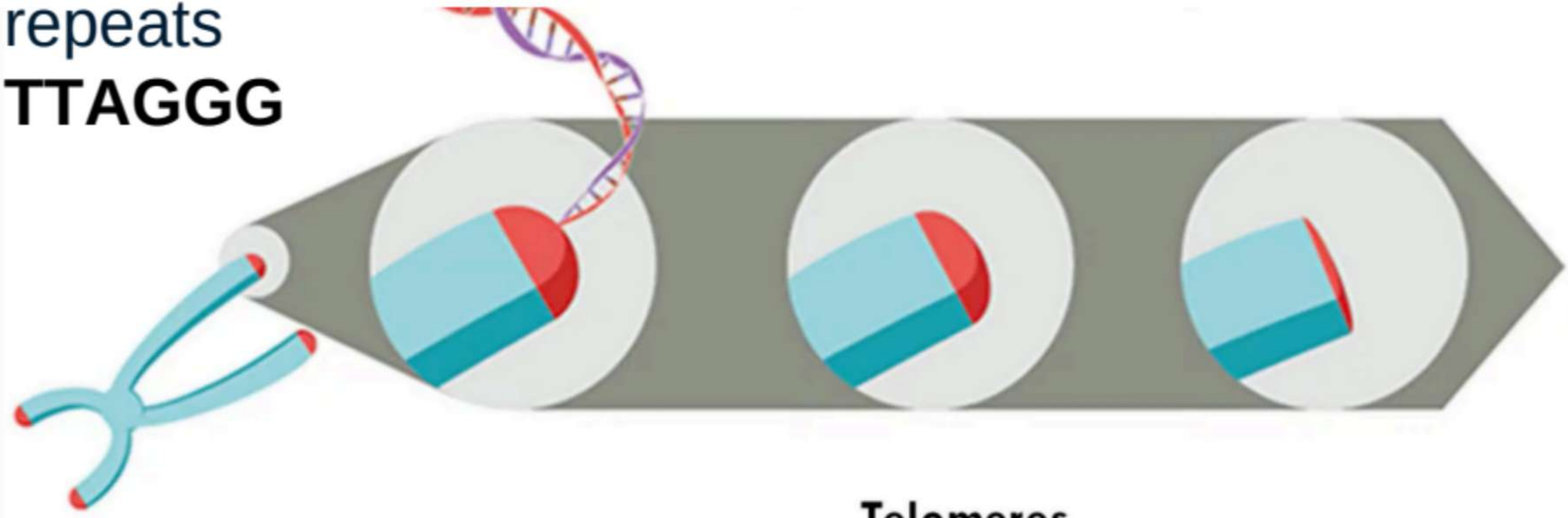


telomeric DNA repeats

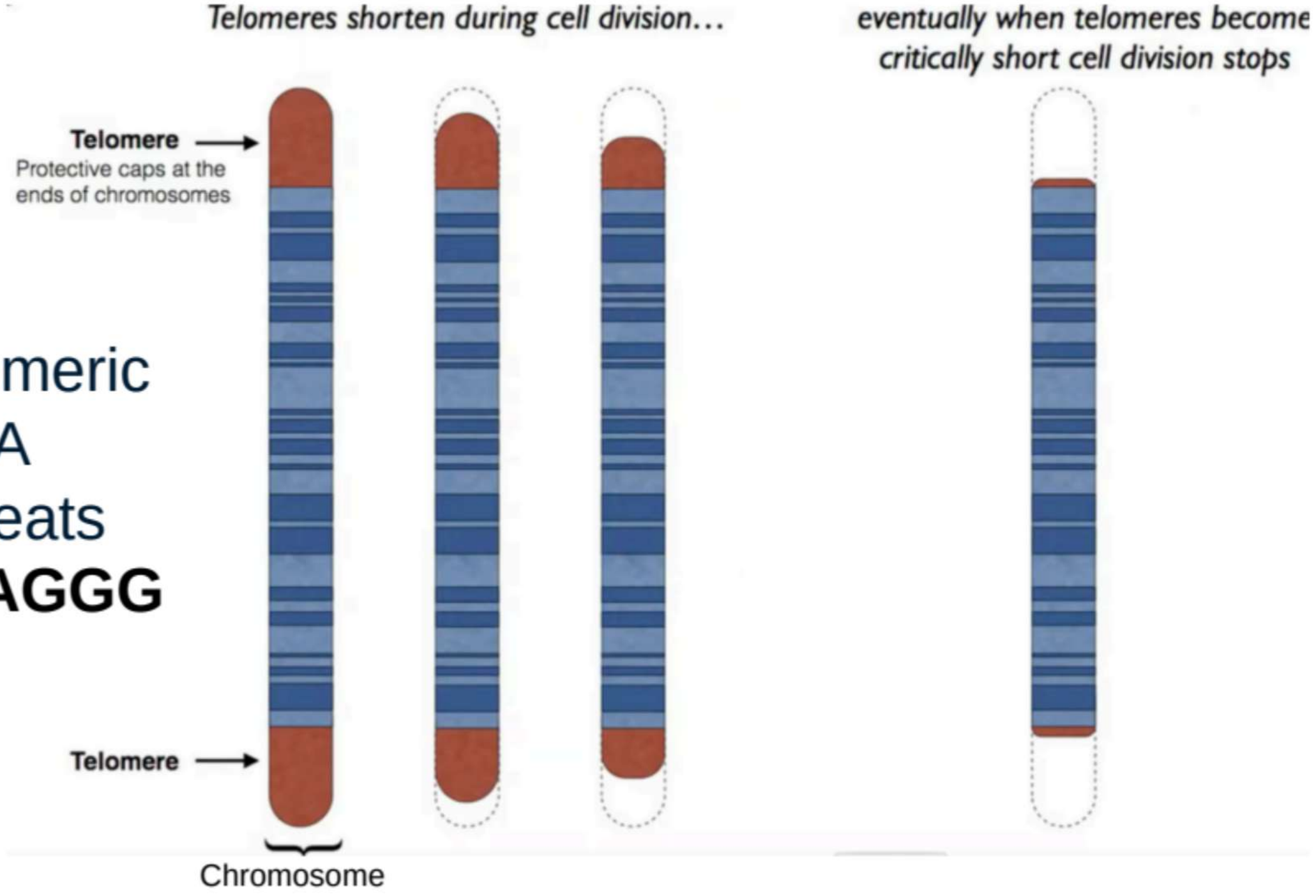
telomeric DNA repeats **TTAGGG**



telomeric DNA repeats **TTAGGG**



Telomeres shorten as the cell divides



telomeric DNA repeats **TTAGGG**

Point Mutations
Deletion of entire regions
Loss of Heterozygosity

HALLMARK II: GENOME INSTABILITY

- There are 46 chromosomes in a normal cell
- Genome Instability gives rise to tumorigenesis
- Mutations stop the cell cycle
- Tumor suppressor genes (TSG) keep cell cycle in check

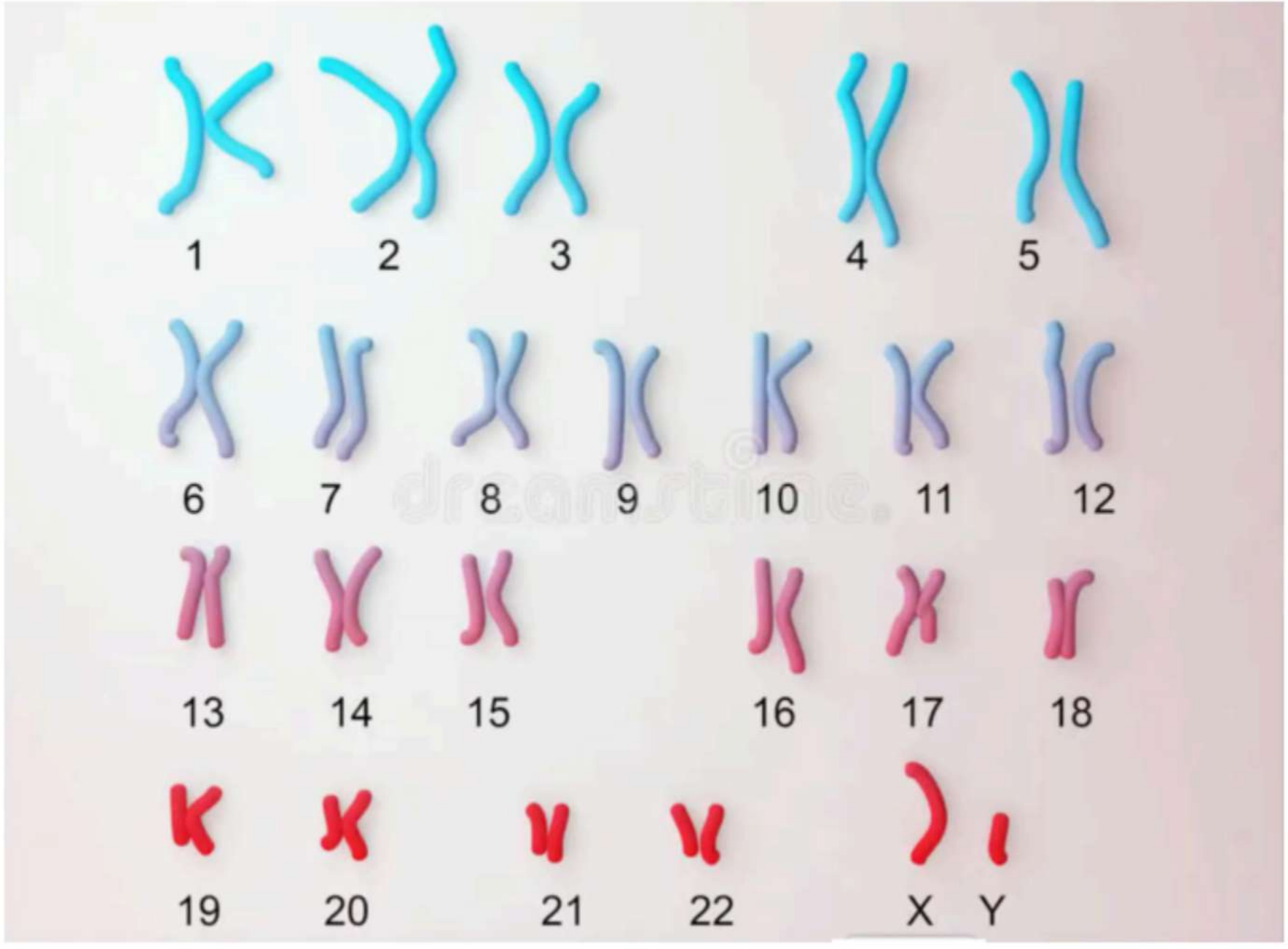


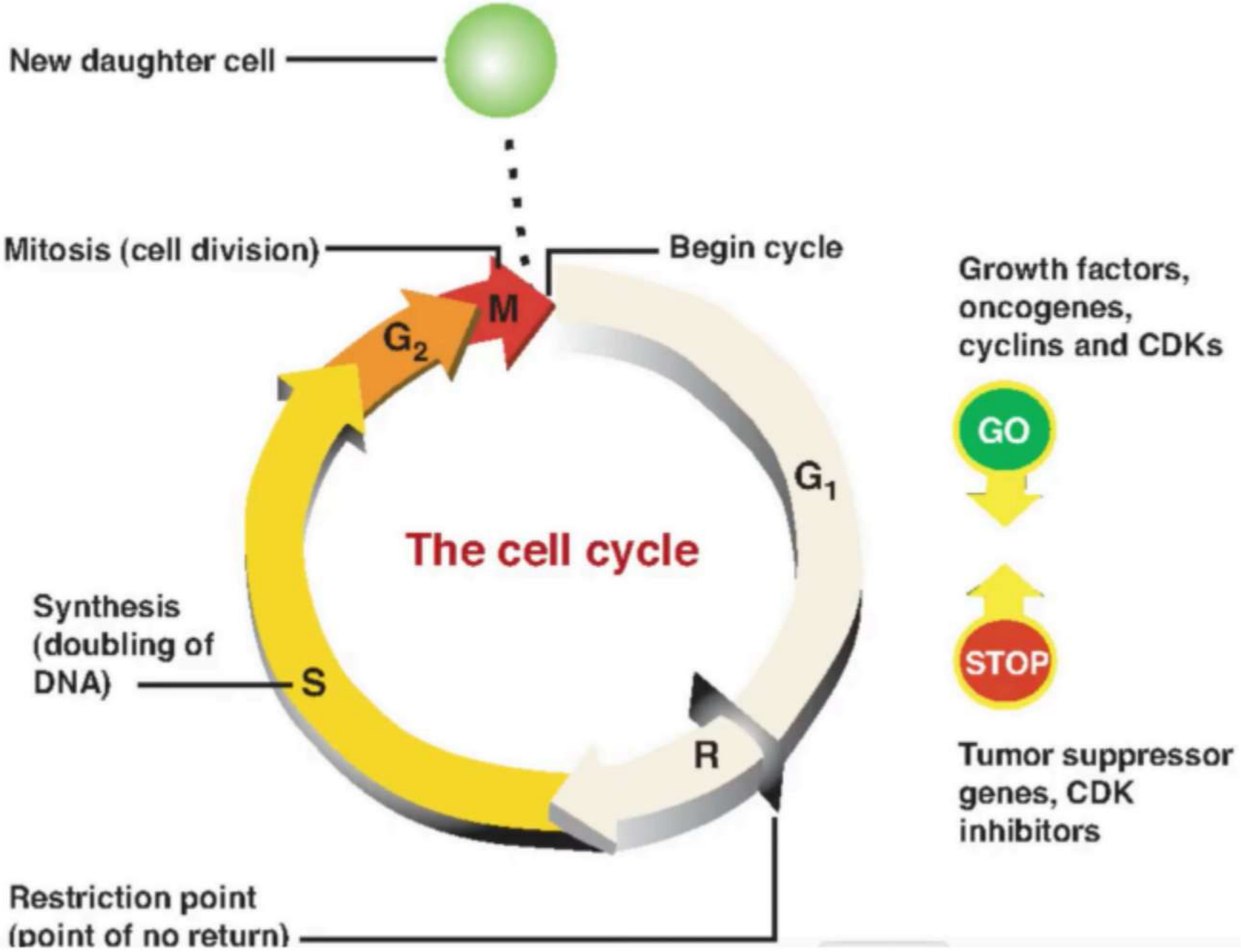
2

Cancer cells have abnormal number of chromosomes per cell

Uncontrolled proliferation occurs due to lost or mutated TSGs

Oncogenes are overly expressed or activate because of mutations


1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
X
Y


New daughter cell
Mitosis (cell division)
Begin cycle
M
G_2
G_1
The cell cycle
Synthesis (doubling of DNA)
S
R
Restriction point (point of no return)
Growth factors, oncogenes, cyclins and CDKs
GO
STOP
Tumor suppressor genes, CDK inhibitors

Point Mutations
Deletion of entire regions
Loss of Heterozygosity

HALLMARK II: GENOME INSTABILITY

- There are 46 chromosomes in a normal cell
- Genome Instability gives rise to tumorigenesis
- Mutations stop the cell cycle
- Tumor suppressor genes (TSG) keep cell cycle in check



2

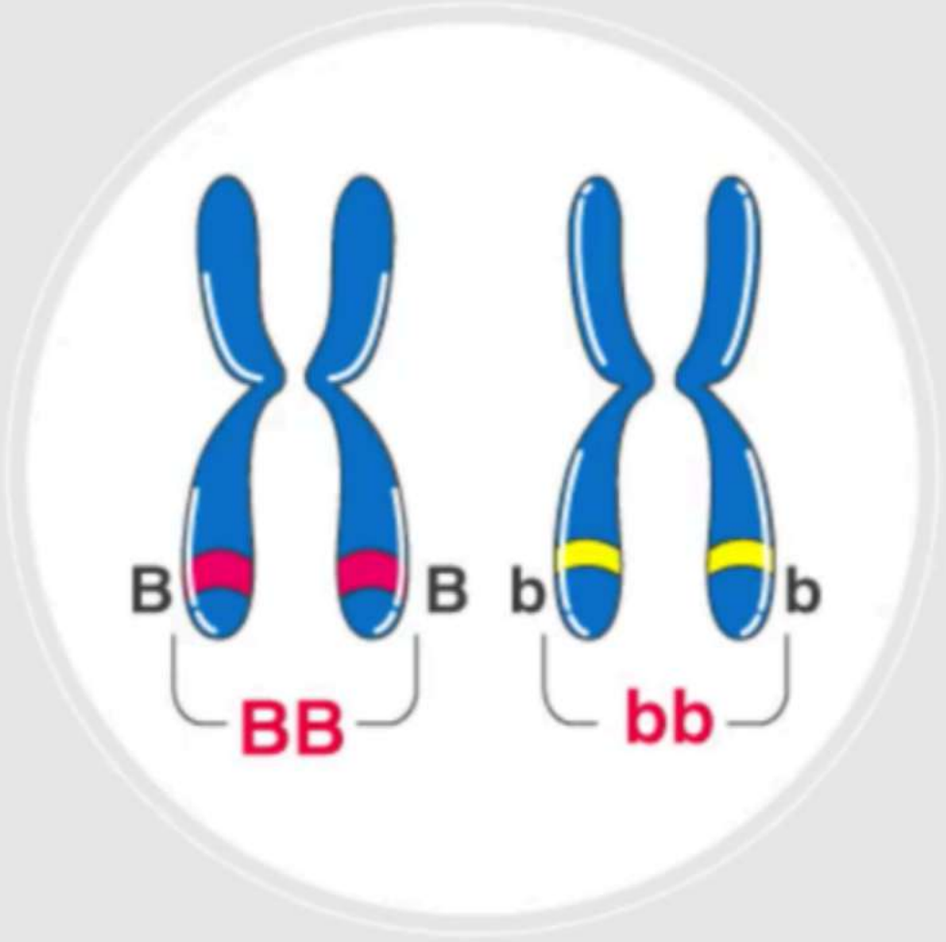
Cancer cells have abnormal number of chromosomes per cell

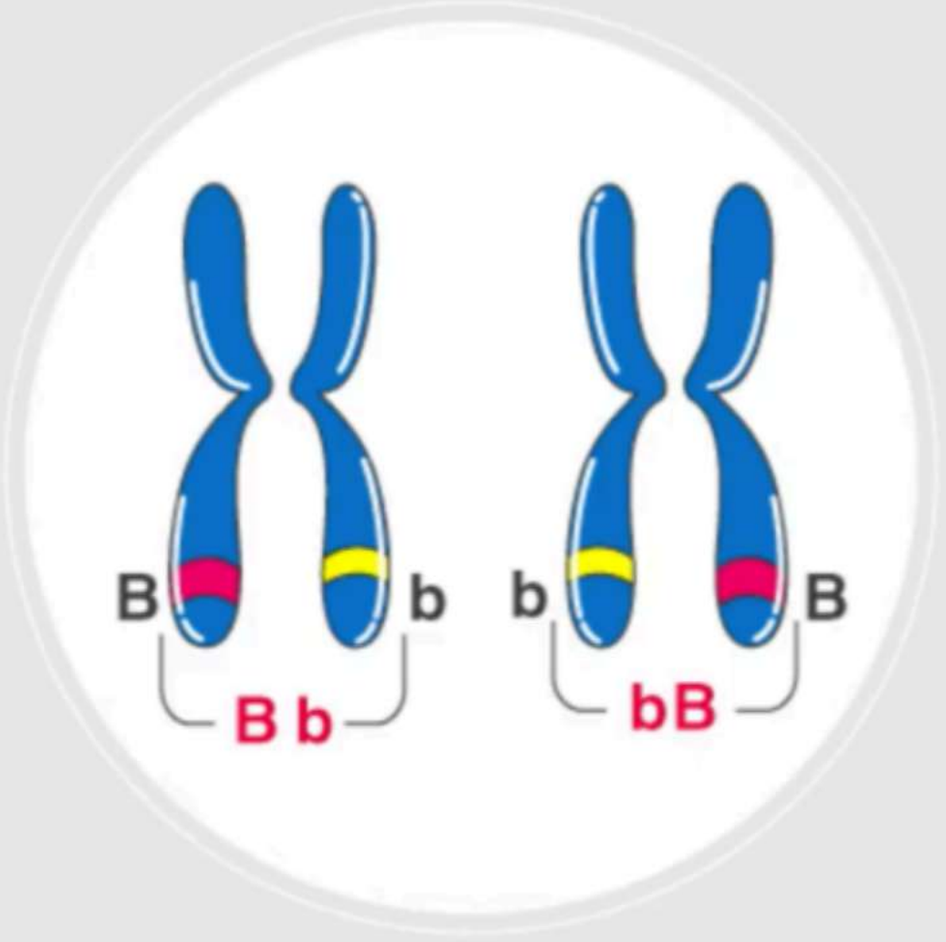
Uncontrolled proliferation occurs due to lost or mutated TSGs

Oncogenes are overly expressed or activate because of mutations

HOMOZYGOUS VS HETEROZYGOUS

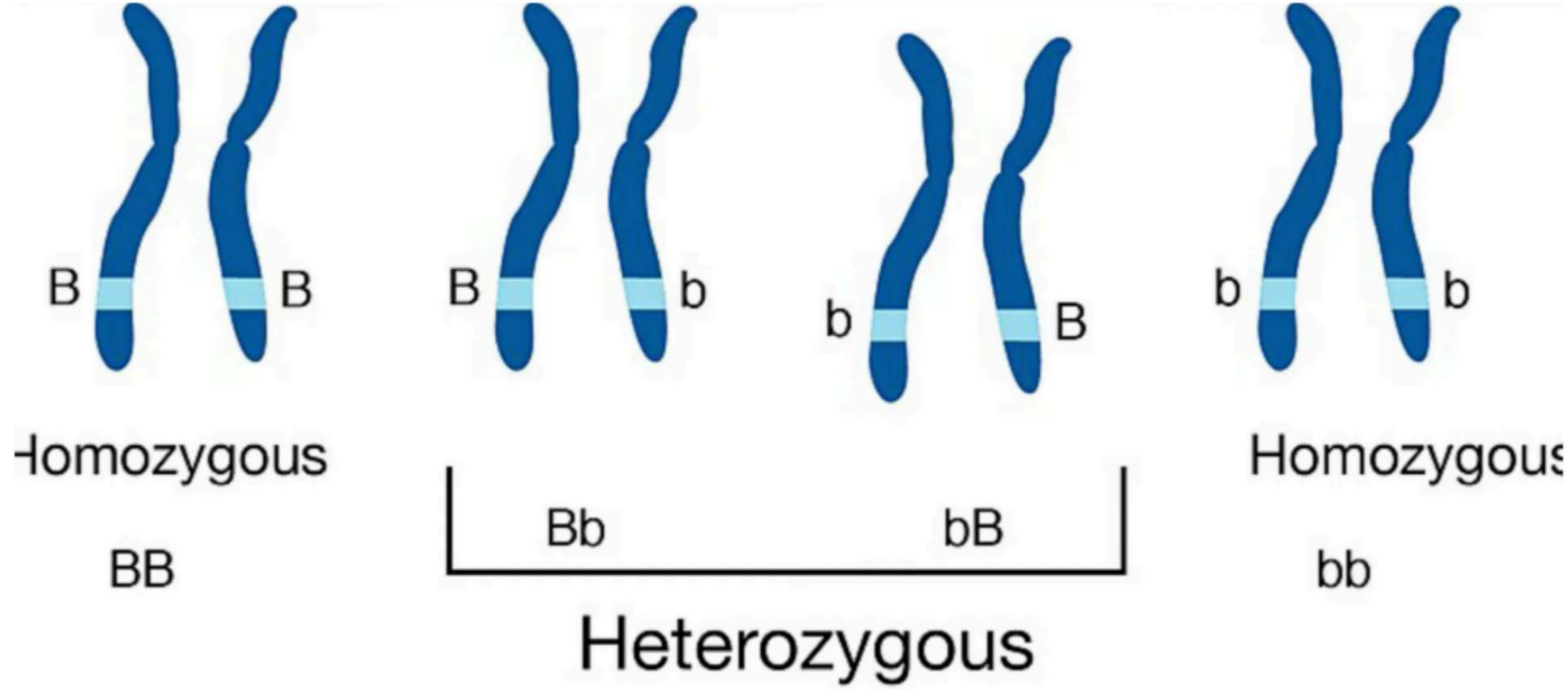
BYJU'S The Learning App

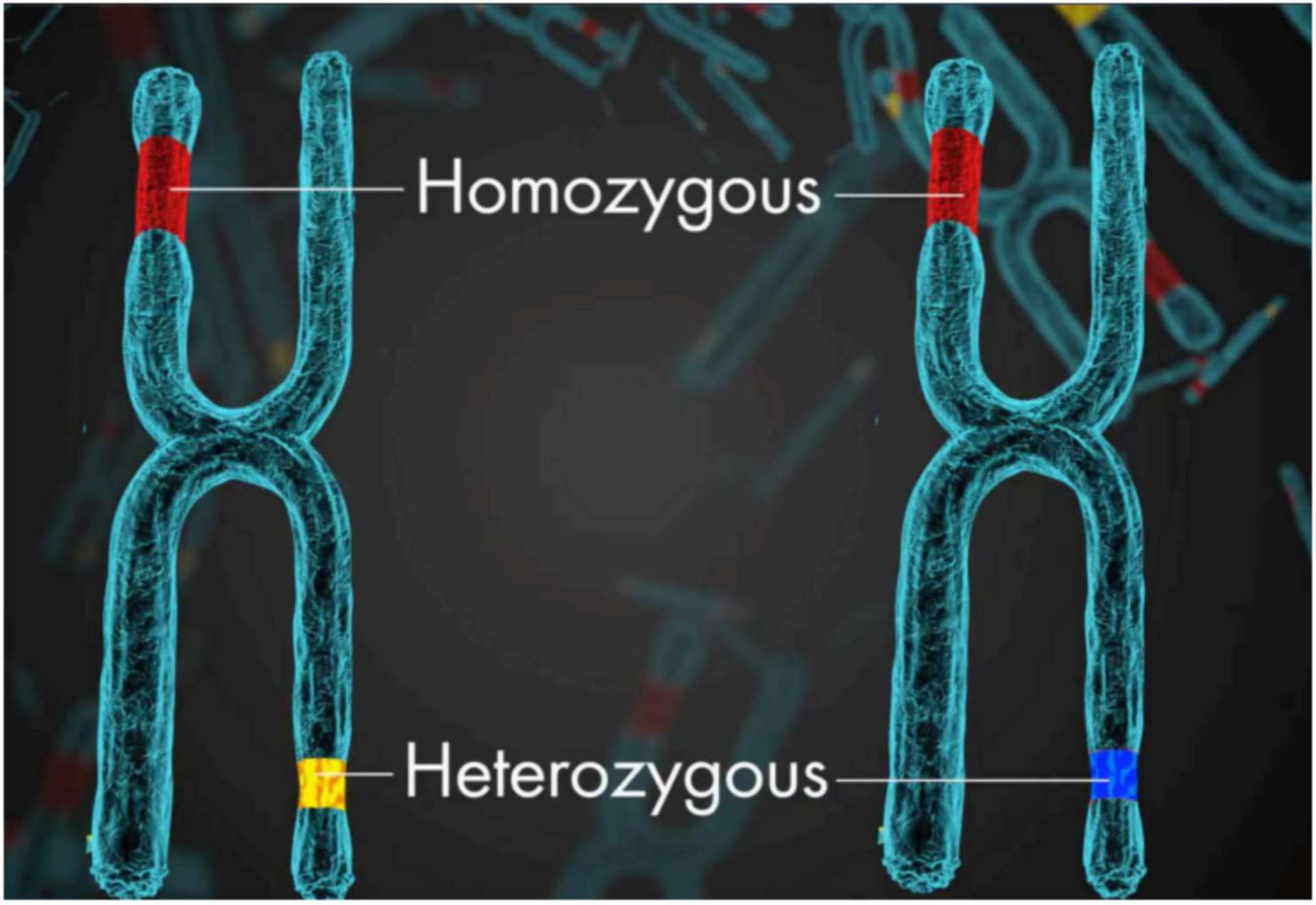




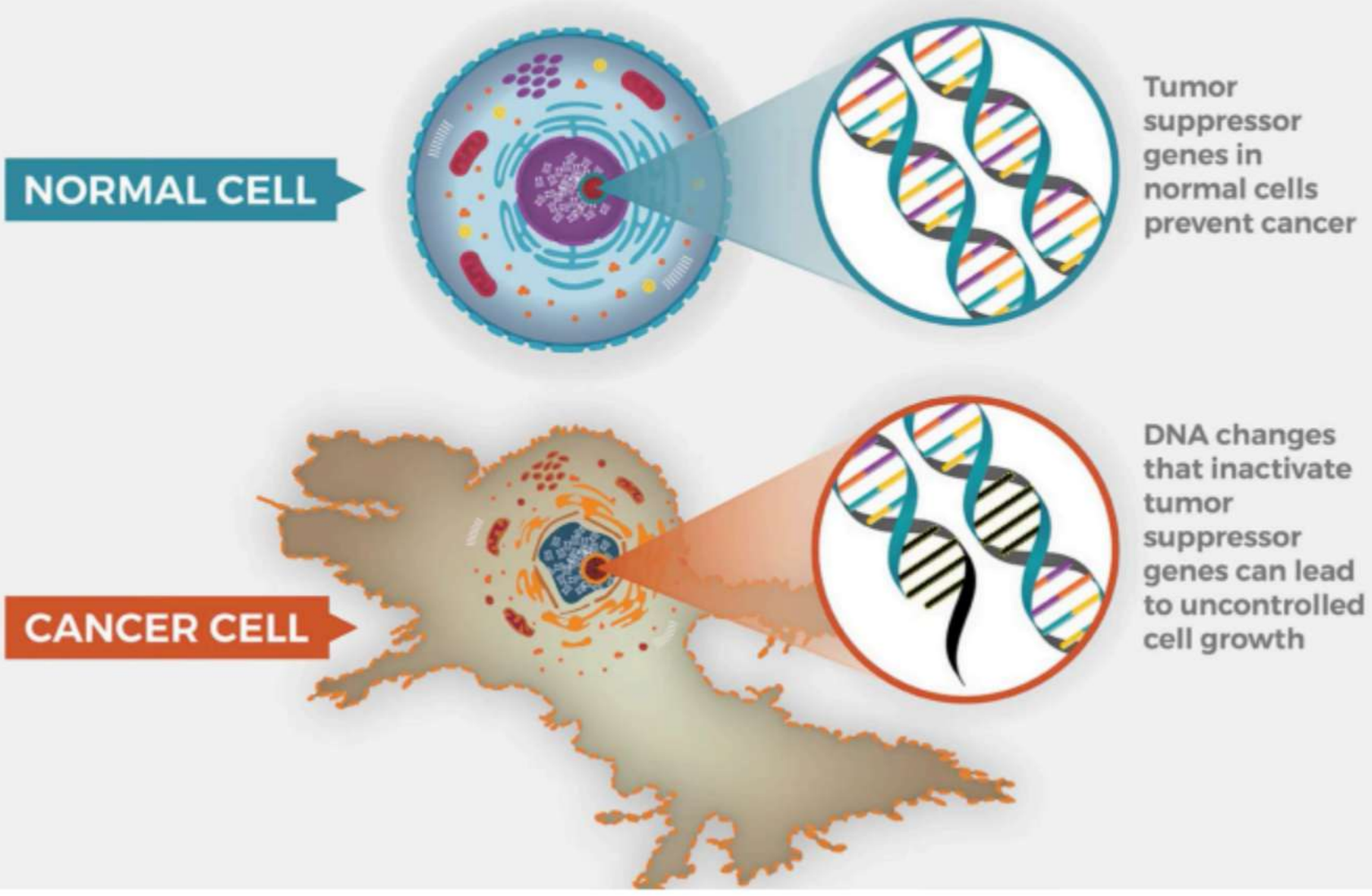
HOMOZYGOUS

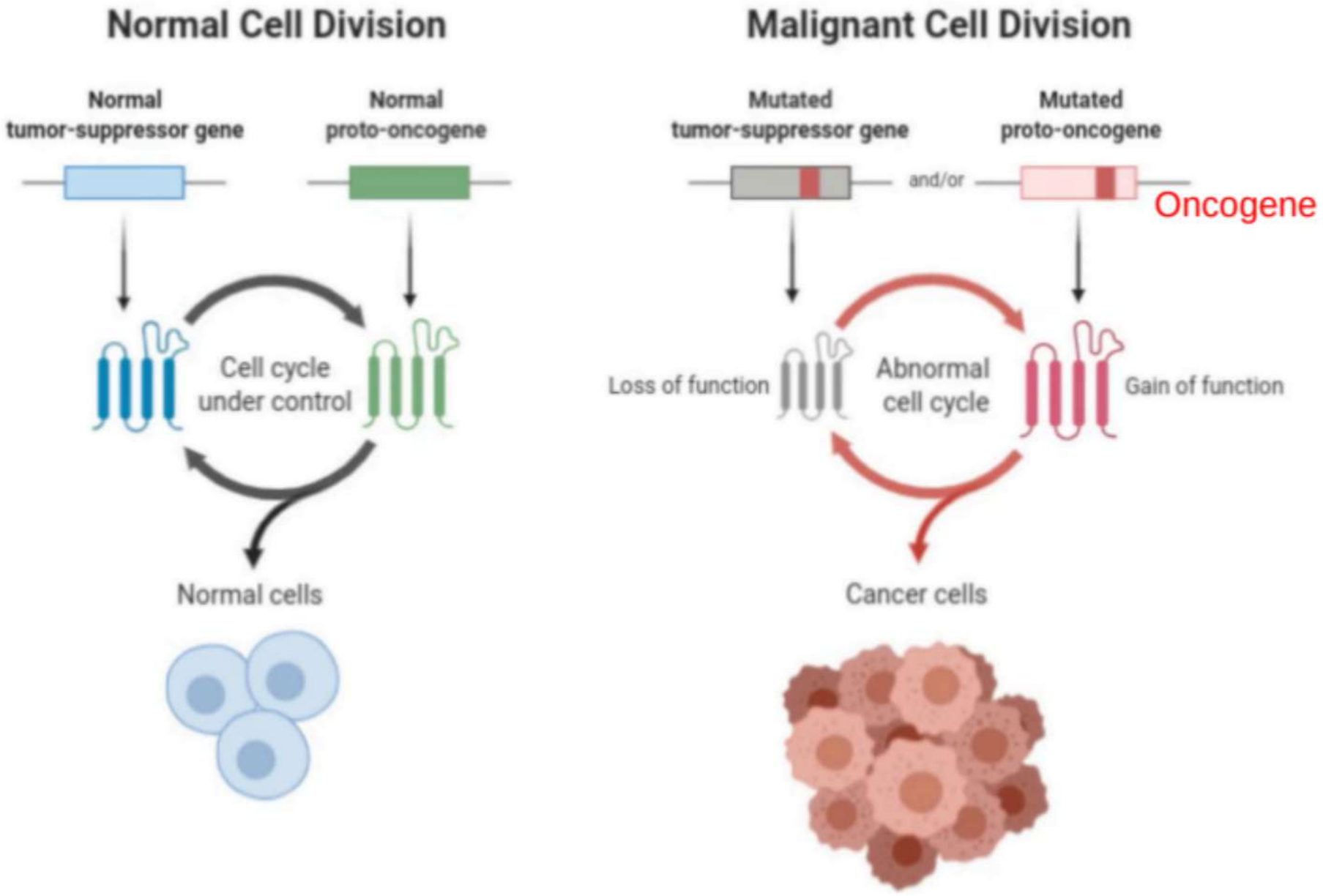
HETEROZYGOUS


B
B
B
b
b
B
b
b
Homozygous
BB
Bb
bB
Heterozygous
Homozygous
bb


Homozygous
Heterozygous

What Are Tumor Suppressor Genes?

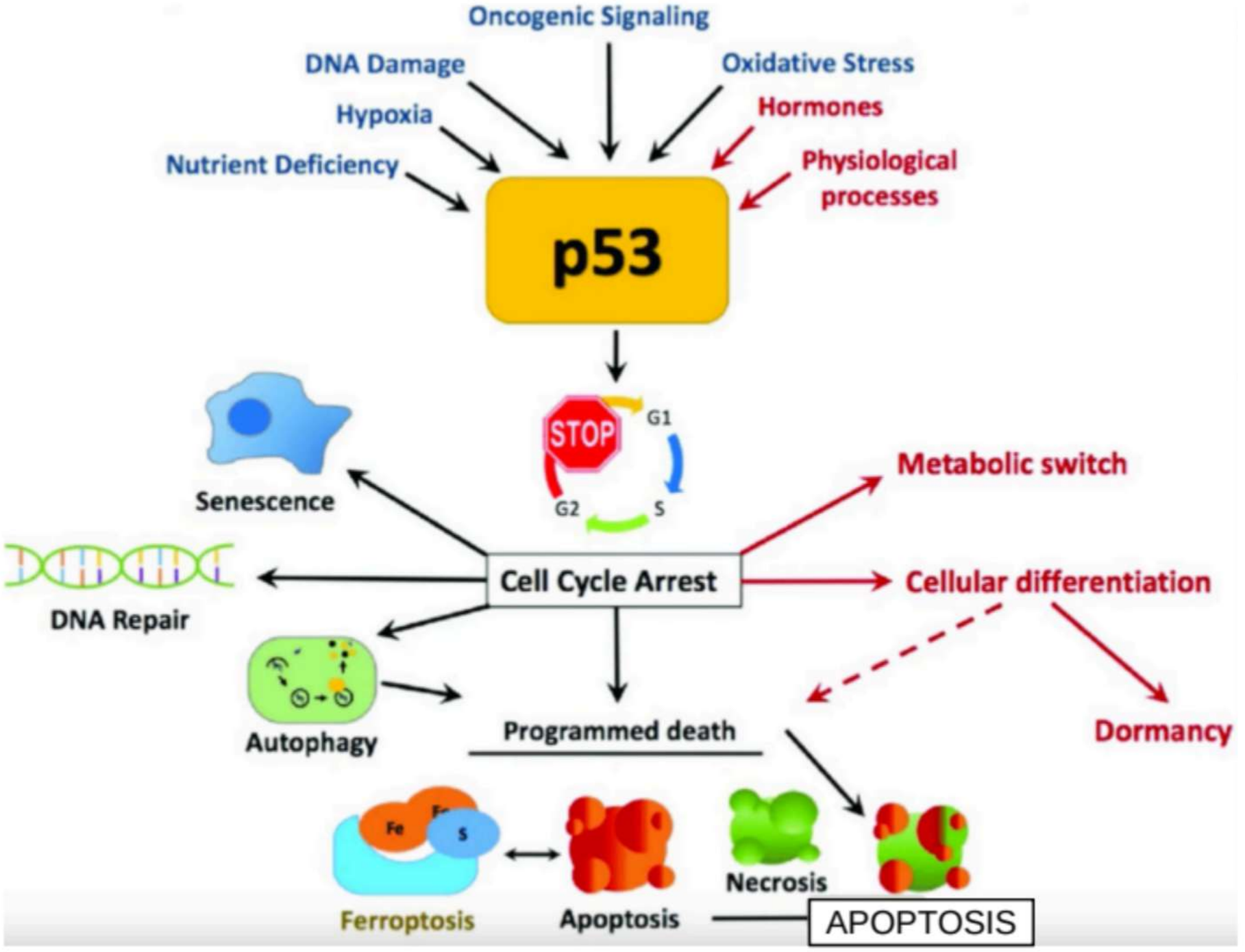



Normal Cell Division
Normal
tumor-suppressor gene
Normal
proto-oncogene
Cell cycle
under control
Normal cells
Malignant Cell Division
Mutated
tumor-suppressor gene
and/or
Mutated
proto-oncogene
Oncogene
Loss of function
Abnormal
cell cycle
Gain of function
Cancer cells

Representative Tumor-Suppressor Genes Implicated in Human Tumors

Tumor-Suppressor Gene	Neoplasm(s)
RBI	Retinoblastoma; osteosarcoma; carcinoma of breast, bladder, and lung
p53	Astrocytoma; carcinoma of breast, colon, and lung; osteosarcoma
WT1	Wilms' tumor
DCC	Carcinoma of colon
NF1	Neurofibromatosis type 1
FAP	Carcinoma of colon
MEN-1	Tumors of parathyroid, pancreas, pituitary, and adrenal cortex

Adapted from Bishop, J. M. 1991. Molecular themes in oncogenesis. *Cell*


Oncogenic Signaling
DNA Damage
Oxidative Stress
Hypoxia
Hormones
Nutrient Deficiency
Physiological processes
p53
STOP
G1
G2
S
Senescence
Metabolic switch
DNA Repair
Cell Cycle Arrest
Cellular differentiation
Autophagy
Programmed death
Dormancy
Fe
S
Necrosis
Ferroptosis
Apoptosis
APOPTOSIS

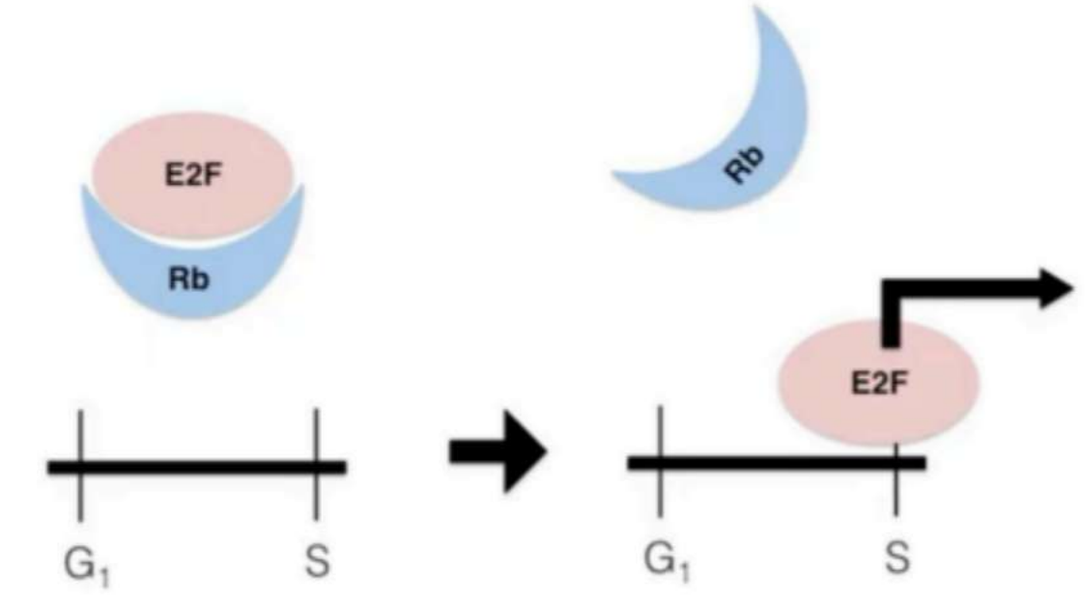
EVASION OF GROWTH SUPRESSOR SIGNALS

HALLMARK
THREE

HALLMARK III: EVASION OF GROWTH SUPPRESSOR SIGNALS

- Anti-growth signals occur in G1 phase
- Cancer cells circumvent (outwit) G1 checkpoint using TSGs for e.g., Rb and p53
- Rb inhibits normal cells passing through restriction points in G1 phase
- Mutated Rb allows cancer cells to proliferate

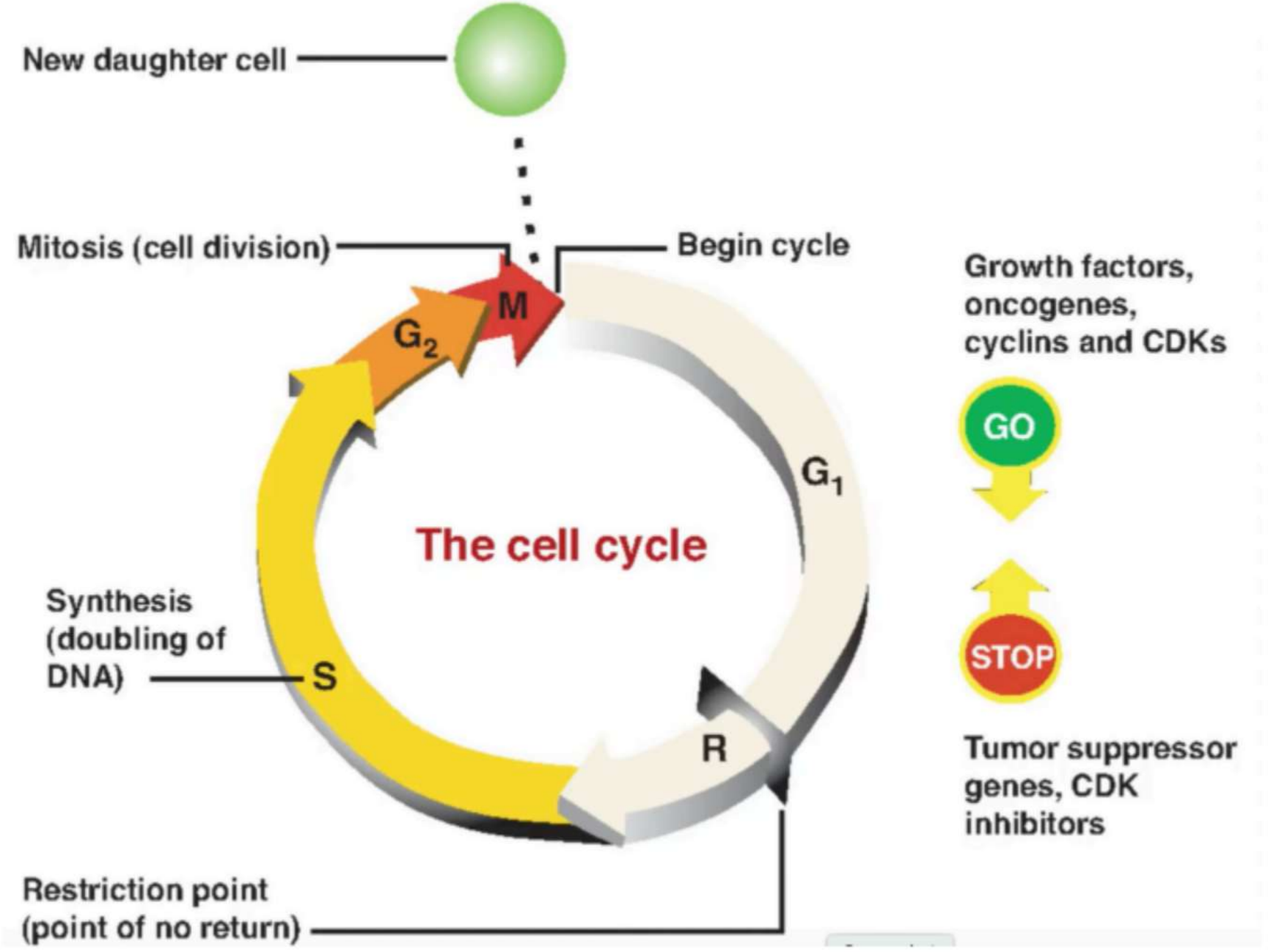
3



Inactive: Rb bound to E2F, **preventing entry** to S phase

Active: (no Rb bound) E2F acts as a transcription factor, **entering** S phase

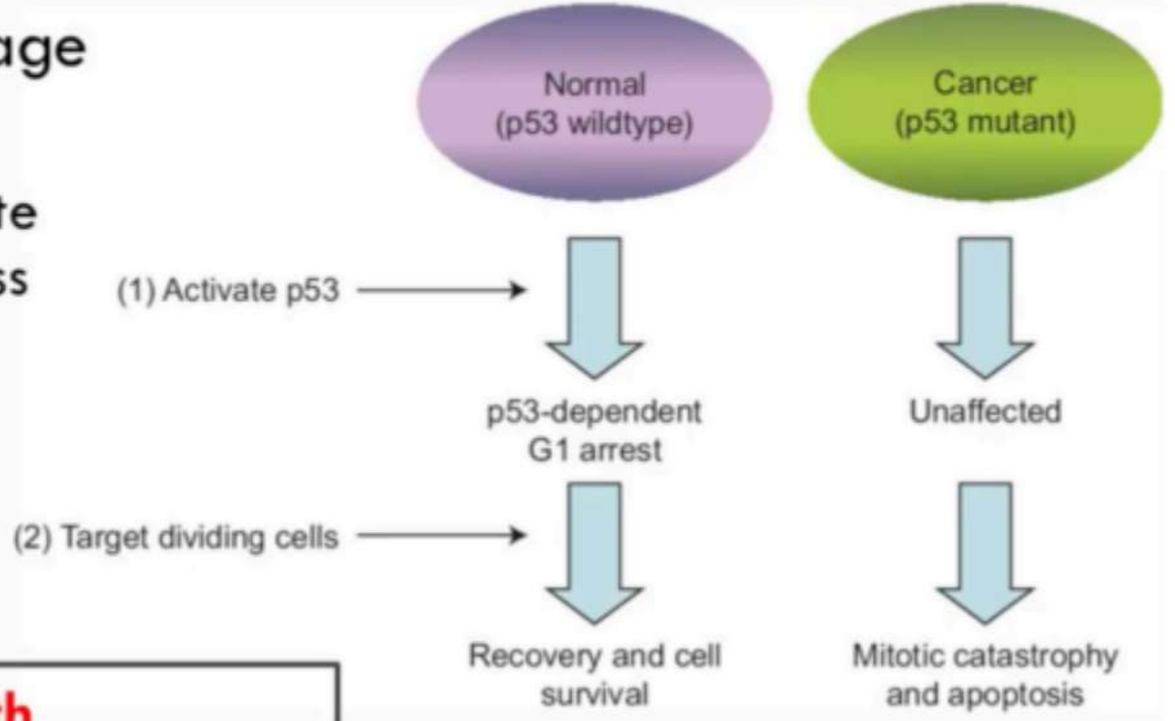
Cell Cycle - Biochemistry - Medbullets Step 1


New daughter cell
Mitosis (cell division)
Begin cycle
M
G_2
G_1
The cell cycle
Synthesis
(doubling of
DNA)
S
R
Restriction point
(point of no return)
Growth factors,
oncogenes,
cyclins and CDKs
GO
STOP
Tumor suppressor
genes, CDK
inhibitors

HALLMARK III: EVASION OF GROWTH SUPPRESSOR SIGNALS

p53 induces apoptosis on DNA damage

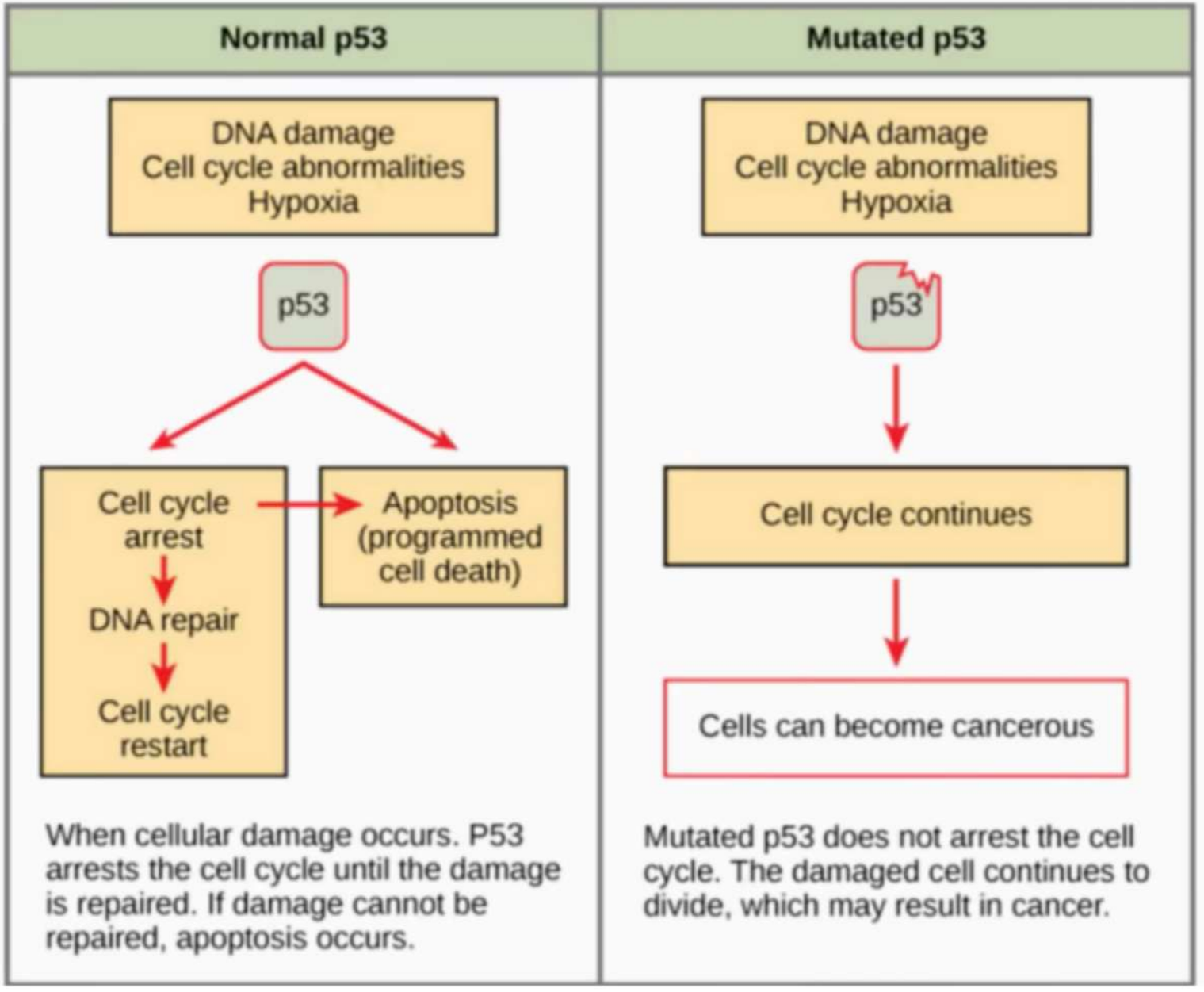
- Mutated p53 allows the cells to proliferate despite DNA damage and oxidative stress



The principle of cyclotherapy to exploit p53 mutation in tumor cells.

3

Cancers cells have no effect of growth suppressor signals due to genetic mutations (#2)


Normal p53
DNA damage
Cell cycle abnormalities
Hypoxia
p53
Cell cycle arrest
DNA repair
Cell cycle restart
Apoptosis (programmed cell death)
When cellular damage occurs. P53 arrests the cell cycle until the damage is repaired. If damage cannot be repaired, apoptosis occurs.
Mutated p53
DNA damage
Cell cycle abnormalities
Hypoxia
p53
Cell cycle continues
Cells can become cancerous
Mutated p53 does not arrest the cell cycle. The damaged cell continues to divide, which may result in cancer.

RESISTANCE TO CELLULAR DEATH

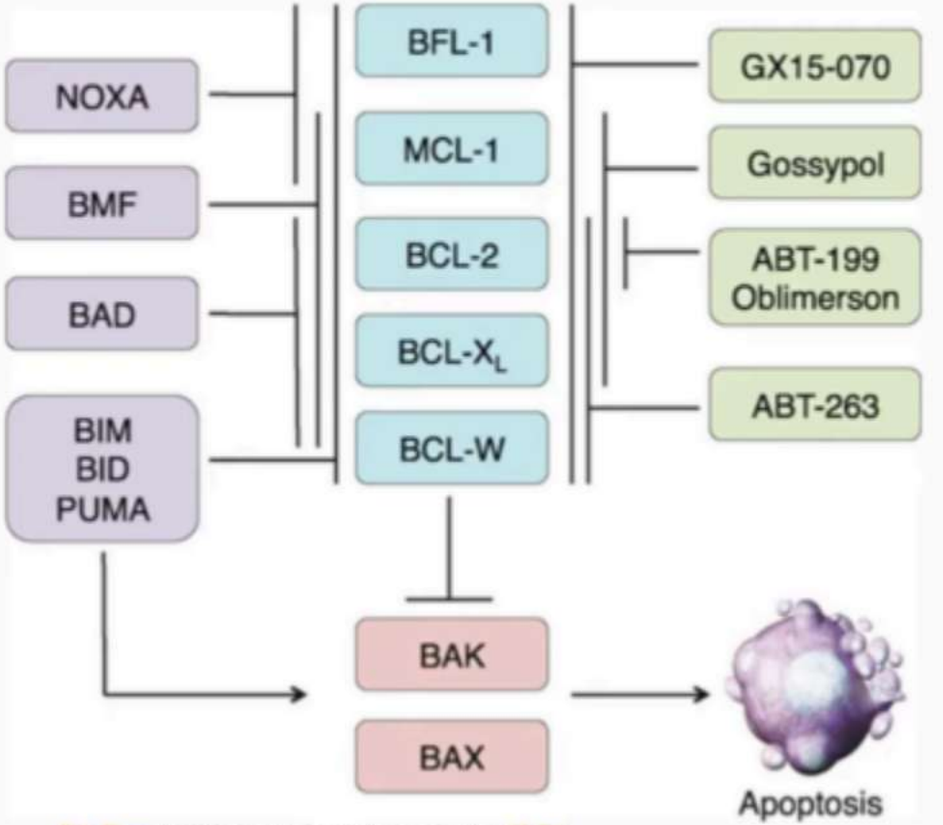
HALLMARK FOUR

Screenshot

HALLMARK IV: RESISTANCE TO CELLULAR DEATH

Irreparable DNA damage induces apoptosis

Cancer cells upregulate pro-survival proteins (for e.g. The BCL-2 Family)



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4

SUSTAINED PROLIFERATION

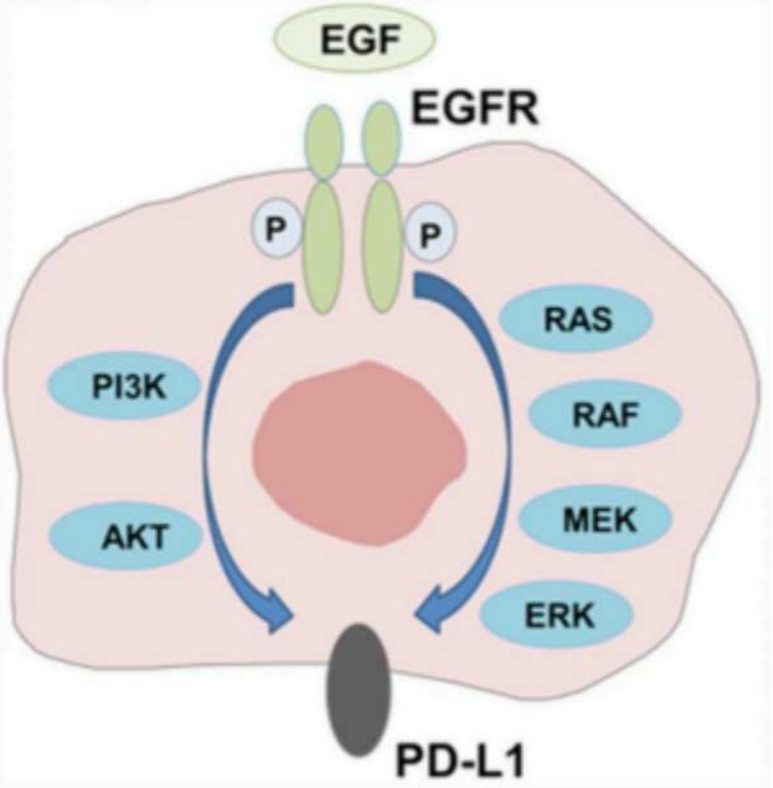
HALLMARK
FIVE

HALLMARK V: SUSTAINED PROLIFERATION

Normal cells have controlled cellular signaling

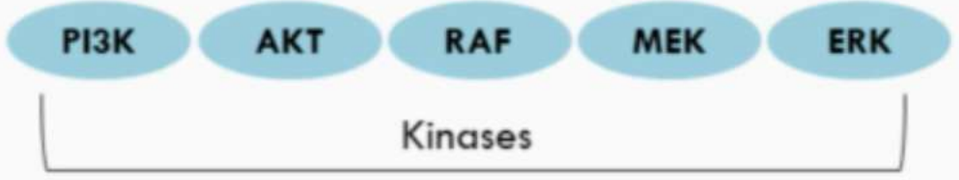
Cancer cells proliferate due to

- Loss of TSGs
- Over expression of oncogenes for e.g., RAS
- Growth Factors (for e.g., EGF) given by healthy cells in the TME



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5



PI = Phospho Inositide 3 Kinase | AKT = Protein Kinase B | RAF = Rapidly Accelerated Fibrosarcoma Sarcoma | MEK = Mitogen Activated Kinase | ERK = Extracellular signal Regulated Kinase

ALTERED METABOLISM

HALLMARK
SIX

HALLMARKS VI: ALTERED METABOLISM

For uncontrolled proliferation (#5), cancer cells need energy

Cells use ATP which they get from

1. Aerobic Metabolism (using Oxygen)
2. Anaerobic Metabolism (without Oxygen)

6



HALLMARKS VI: ALTERED METABOLI

Normal cells use glucose to generate ATP

Cancer cells convert glucose into lactate in t presence of oxygen

6

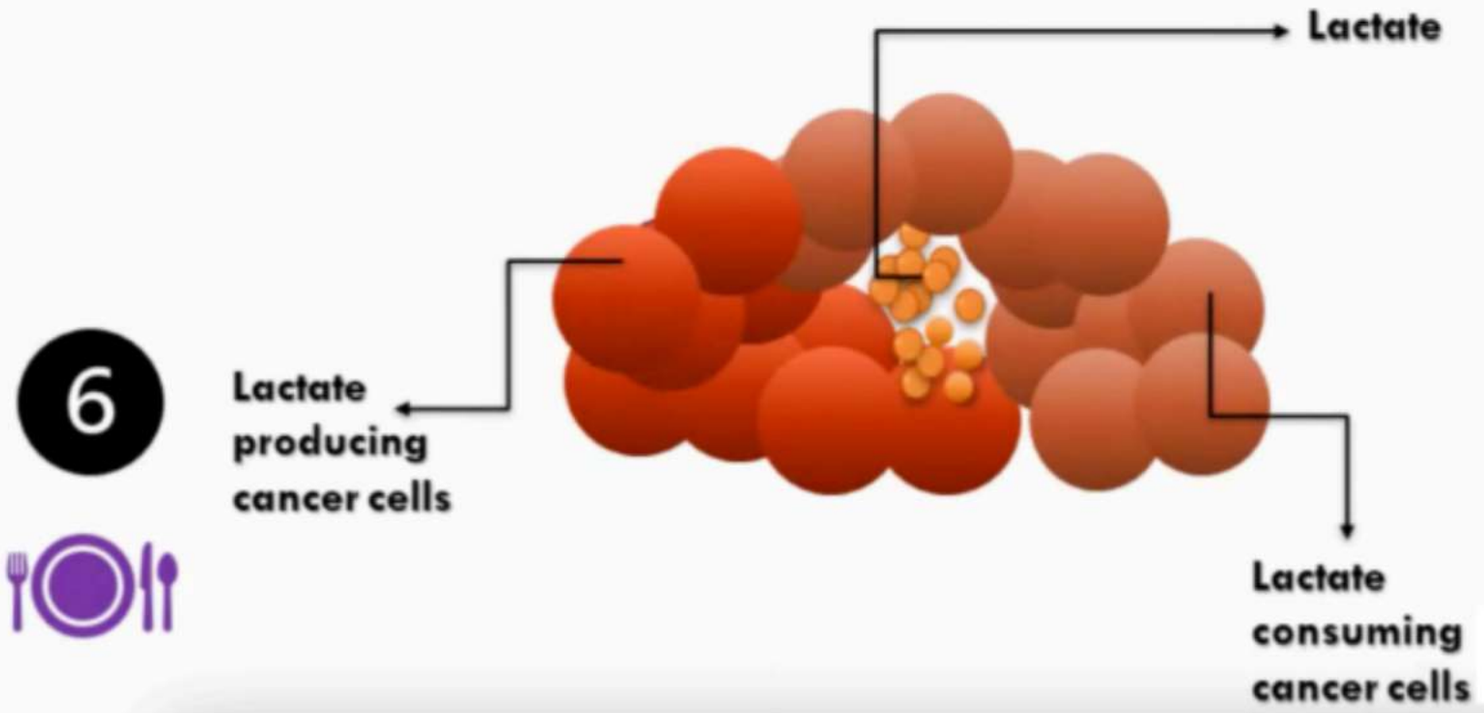
- **This is called the Warburg Effect.**
  - Mutations in enzymes for e.g., IDH-1 and IDH-2



- Warburg effect (aerobic glycolysis)
  - Immunosuppression
  - Lactate makes acidic pH (impair T-cell)
  - Excess ROS promote metastasis (resistance to anoikis) ⊗cell death
  - Remodel TME (promote angiogenesis cancer-associated Fibroblast) and drug resistance
  - New biomass (tumor is formed)

HALLMARKS VI: ALTERED METABOLISM

The lactate produced by cancer cells is fed to other tumor cells along with other metabolites



6

AVOIDING IMMUNE DESTRUCTION

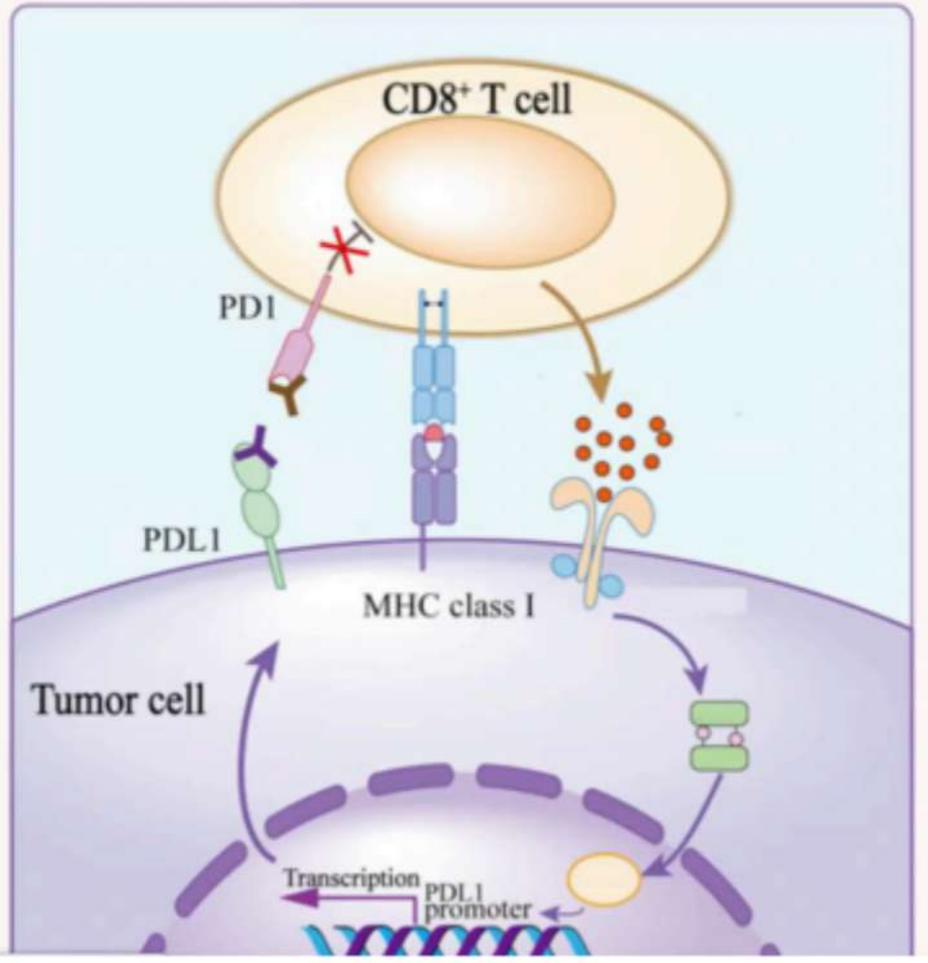
HALLMARK SEVEN

HALLMARK VII: AVOIDING IMMUNE DESTRUCTION

Cancer cells display surface antigens and ligands to avoid immune cells.

Two most common surface markers are

1. Programmed Death Ligand (PDL-1)
2. CD-47



HALLMARK VII: AVOIDING IMMUNE DESTRUCTION

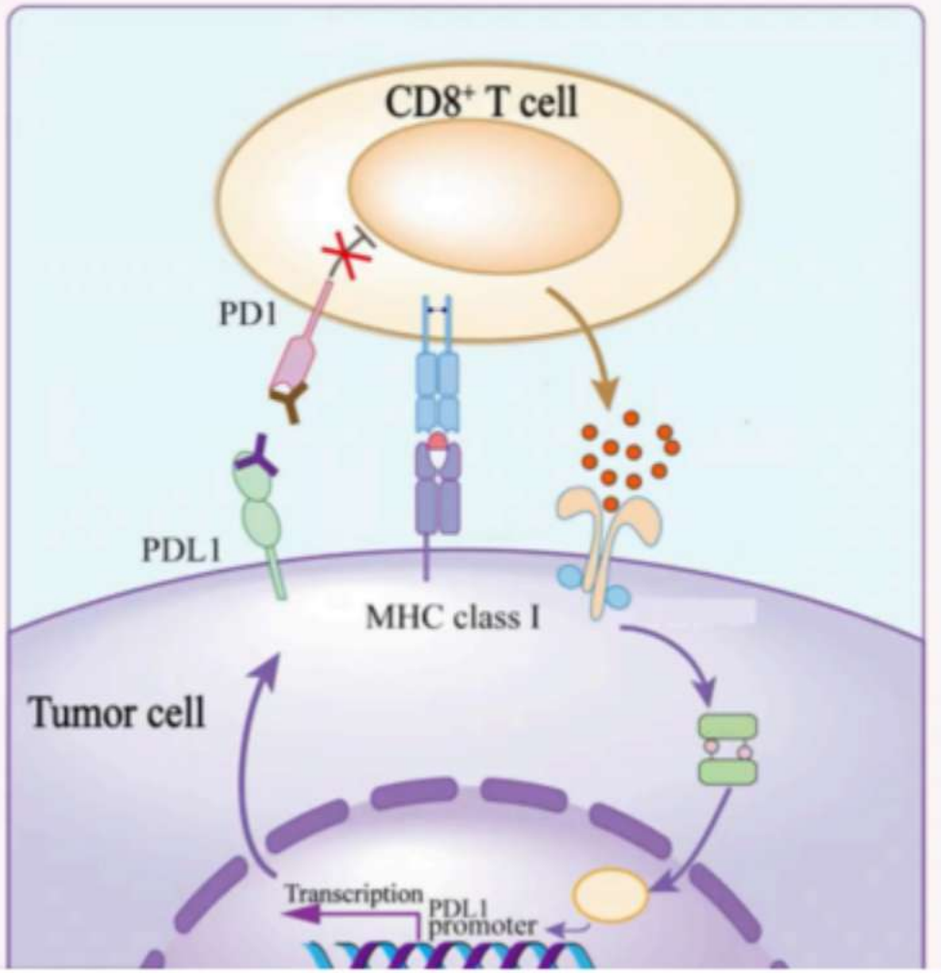
1. Programmed Death Ligand (PDL)

Programmed Death Ligand (PDL1&2) suppresses the immune system in:

a. Pregnancy

b. Transplantations

c. Autoimmune diseases etc.

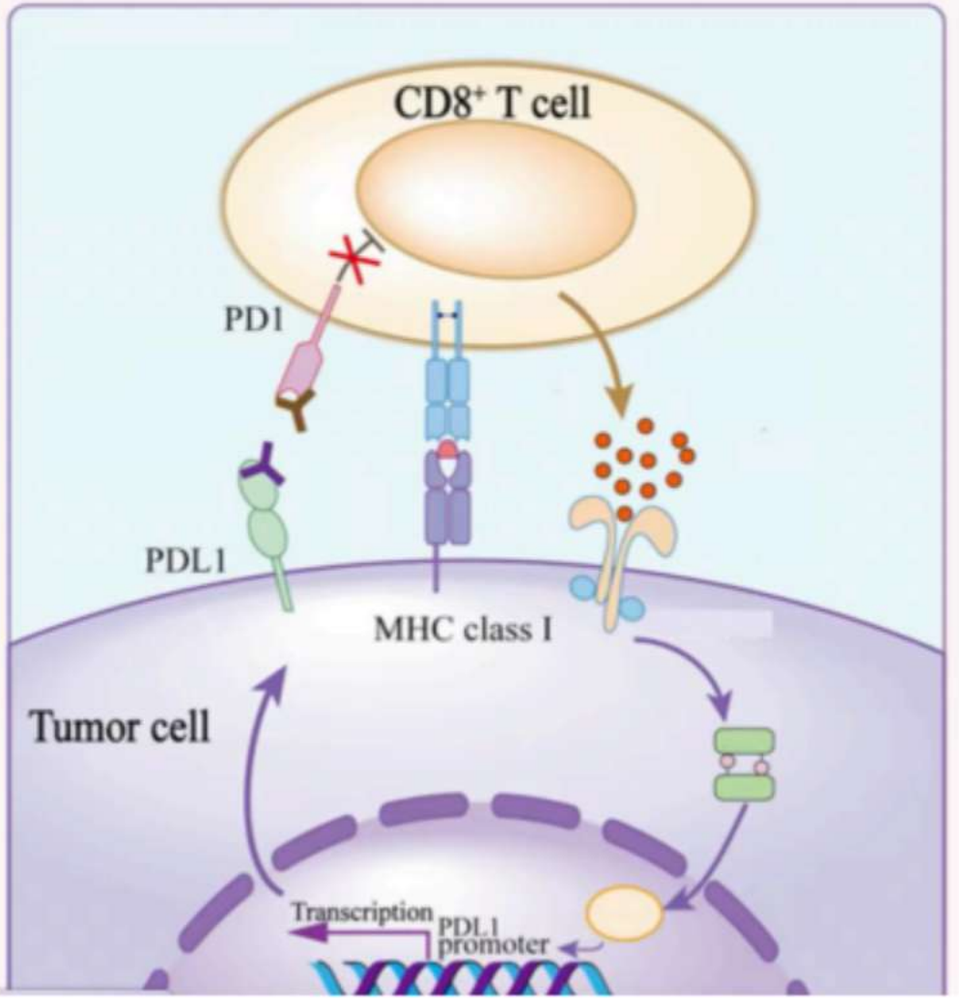


HALLMARK VII: AVOIDING IMMUNE DESTRUCTION

1. Programmed Death Ligand (PDL)

T-cells have a receptor PD-1 that binds with PD-L1 and prevents autoimmunity by stopping the T-cells from proliferation.

Cancer cells have an increased expression of PD-L1



HALLMARK VII: AVOIDING IMMUNE DESTRUCTION

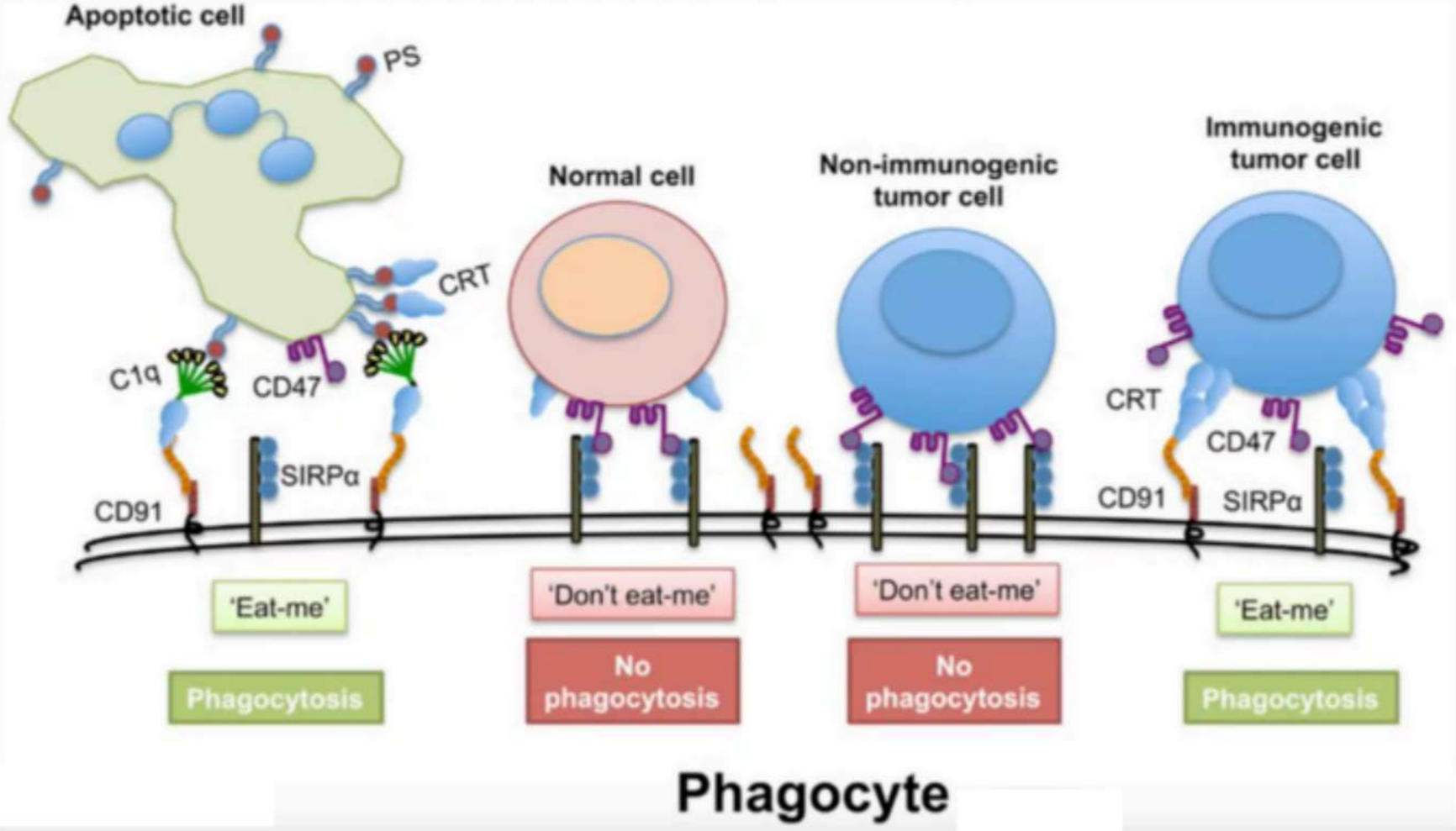
2. **CD-47**

Cancer cells have CD-47 antigens on their surface that stop the macrophages from eating them.

CD-47 gives a DON'T EAT ME signal which blocks phagocytosis



HALLMARK VII: AVOIDING IMMUNE DESTRUCTION

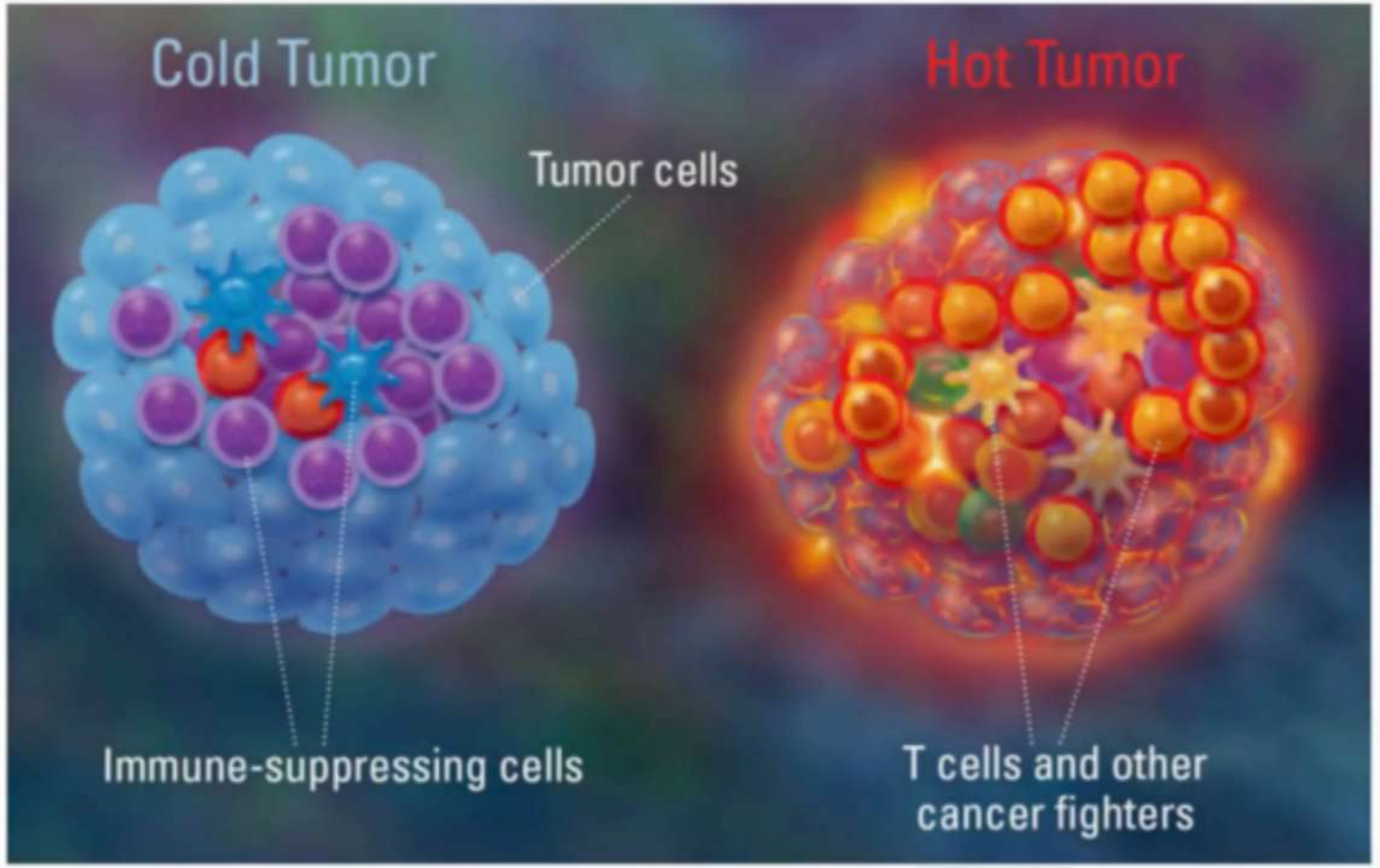


7

TUMOR PROMOTING INFLAMMATION

HALLMARK EIGHT

HALLMARK VIII: TUMOR PROMOTING INFLAMMATION



HALLMARK VIII: TUMOR PROMOTING INFLAMMATION

The TME is infiltrated with innate and adaptive immune cells

Tumors mimic inflammation and categorized in two ways

Hot tumors

- Immune cells activated
- Rapid differentiation

Cold tumors

- Immune cells suppressed
- Slow differentiation



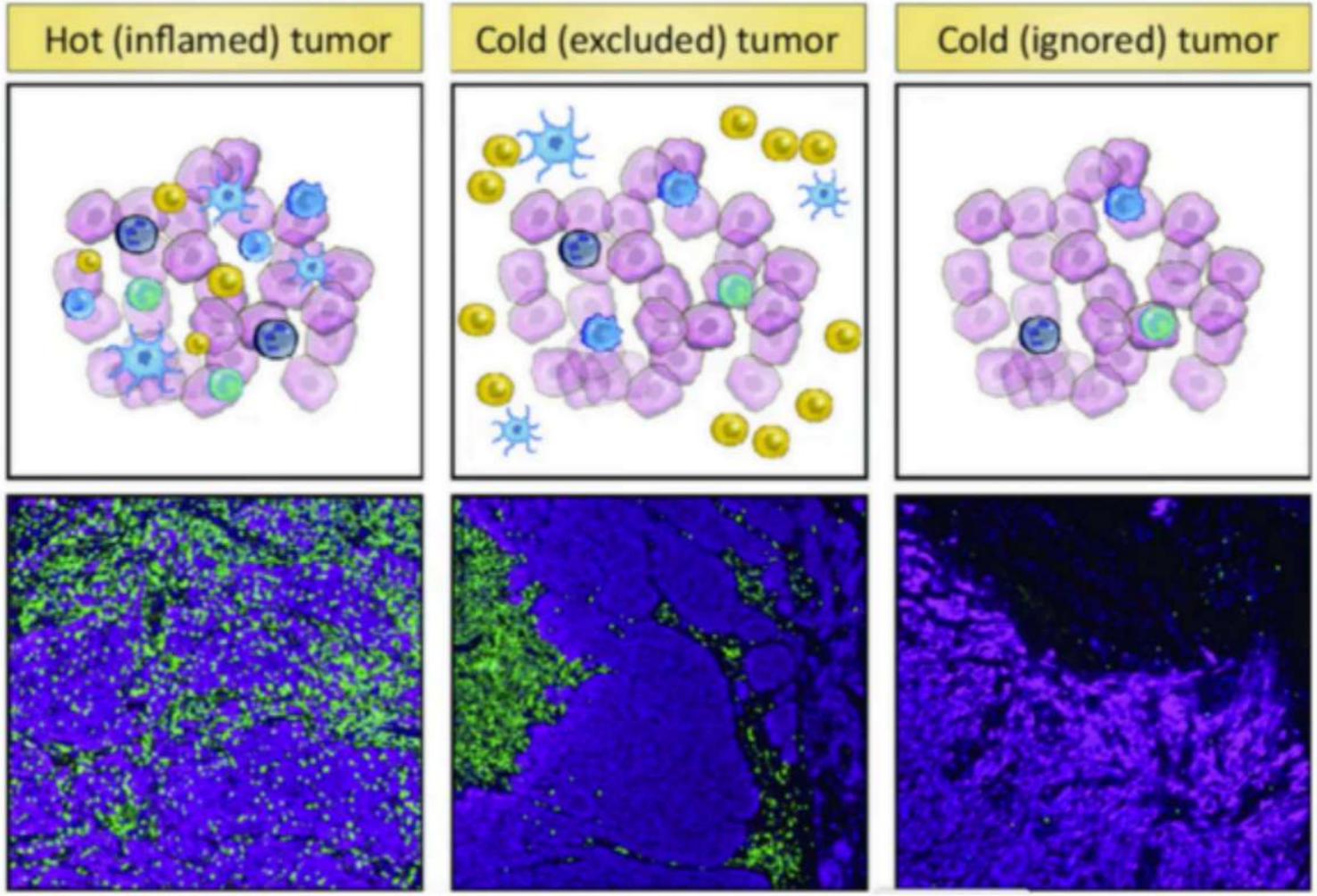
HALLMARK VIII: TUMOR PROMOTING INFLAMMATION

Tumor-associated inflammation:

1. Accelerates tumor development
2. Increases cell survival rate by increasing mutations
3. Enables cell motility and angiogenesis
4. Allows cells to undergo EMT for metastasis



HALLMARK VIII: TUMOR PROMOTING INFLAMMATION



INDUCING ANGIOGENESIS

HALLMARK NINE

HALLMARK IX: ANGIOGENESIS INDUCTION

The formation of new blood vessels is critical for sustained growth and metastasis

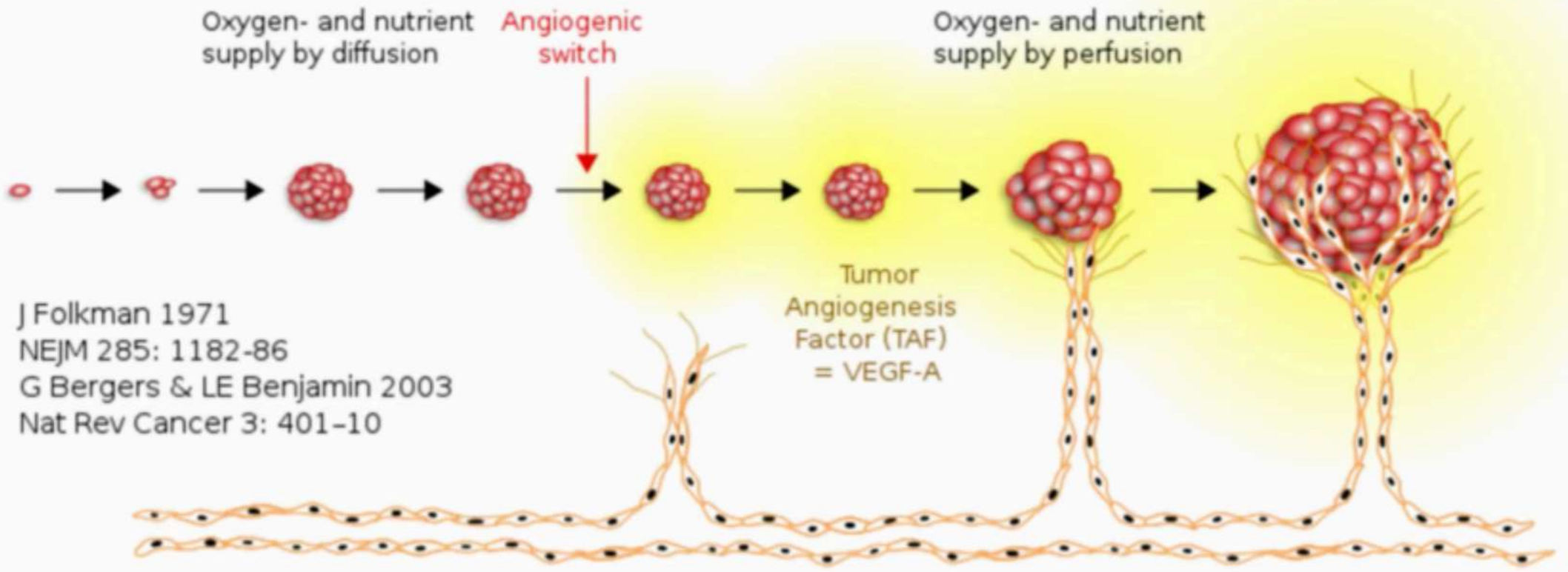
Cancer cells use pro-angiogenic factors to make new blood vessels

- VEGF is activated to promote endothelial cell growth
- Macrophages can also secrete VEGF



VEGF = VASCULAR ENDOTHELIAL GROWTH FACTOR

HALLMARK IX: ANGIOGENESIS INDUCTION



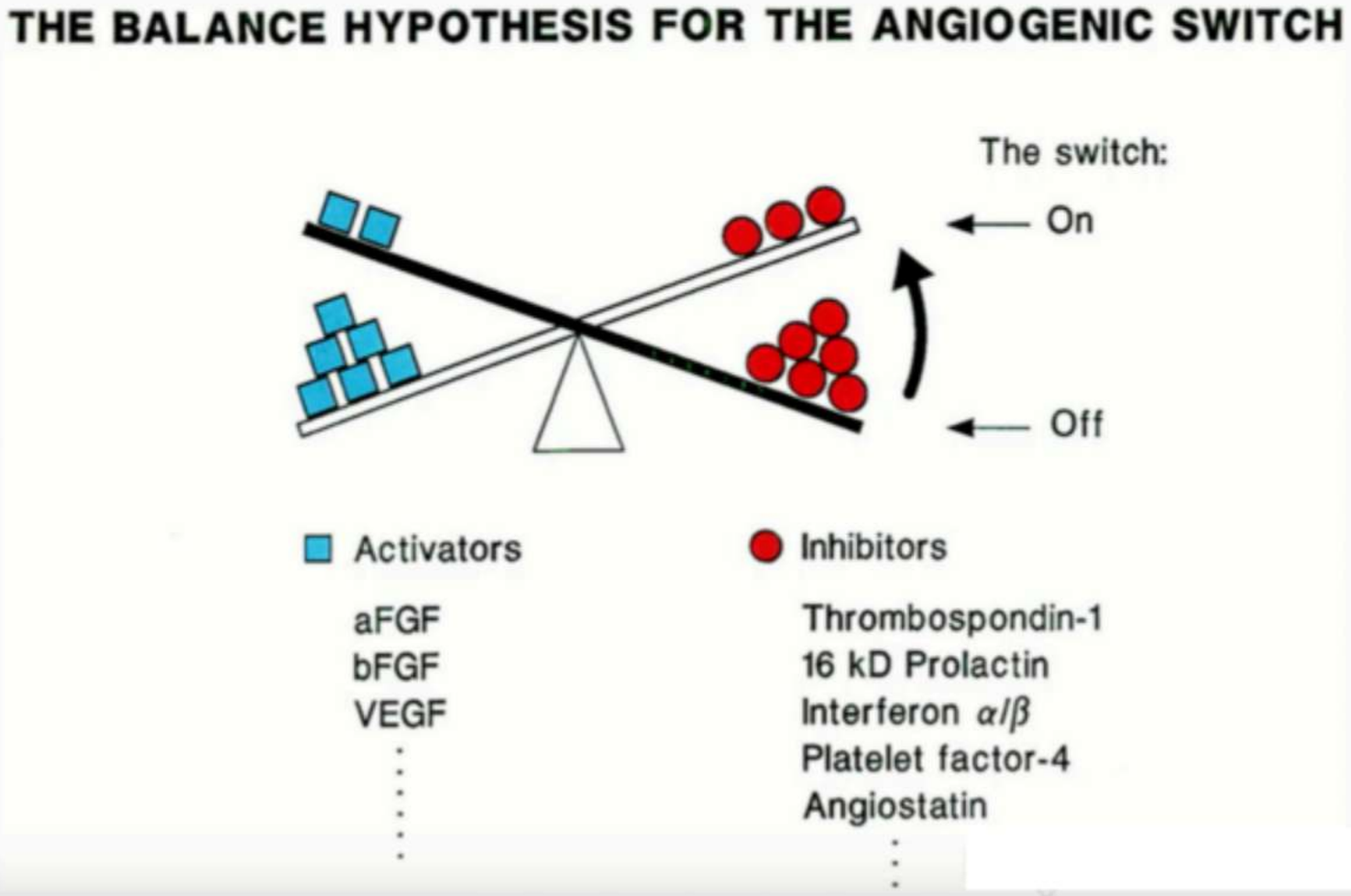
J Folkman 1971
NEJM 285: 1182-86
G Bergers & LE Benjamin 2003
Nat Rev Cancer 3: 401–10

9

Screenshot

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HALLMARK IX: ANGIOGENESIS INDUCTION



HALLMARK X: INVASION & METASTASIS

Prerequisites for a cancer cell to become metastatic:

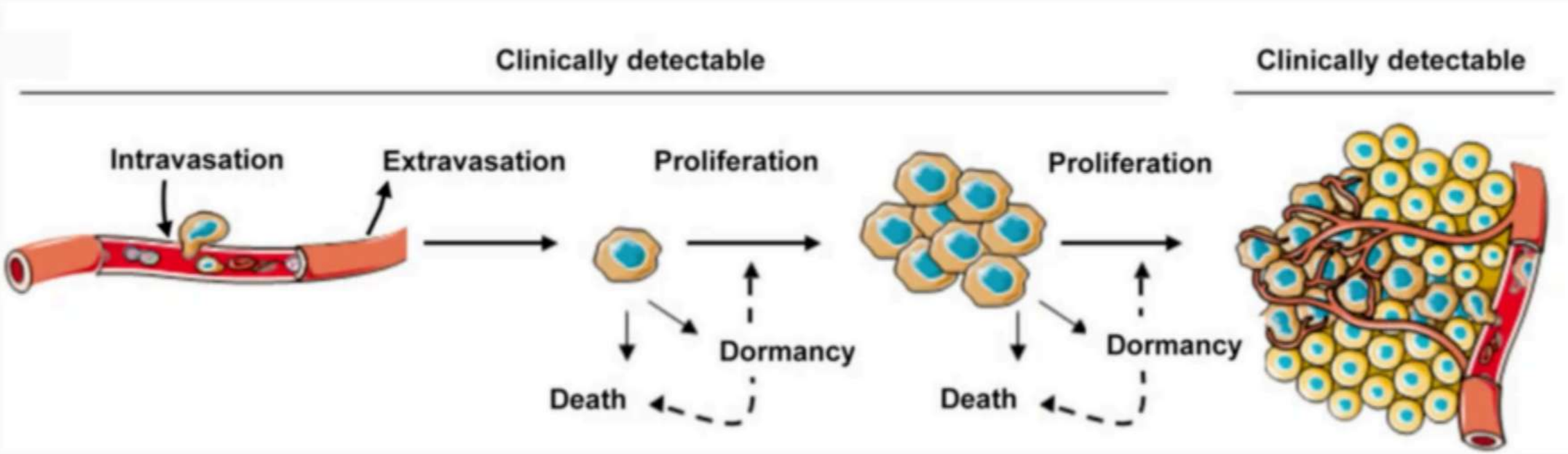
1. Cell-cell and cell-extracellular matrix interactions are altered
2. Structural proteins are mutated or lost (for e.g., integrin loss)
3. Loss of metastasis suppressor genes
4. Recruitment of immune cells
5. Epithelial to Mesenchymal Transition (EMT)



ACTIVATING INVASION & METASTASIS

HALLMARK
TEN

HALLMARK X: INVASION & METASTASIS



10

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KEY STEP 1 | INVASION

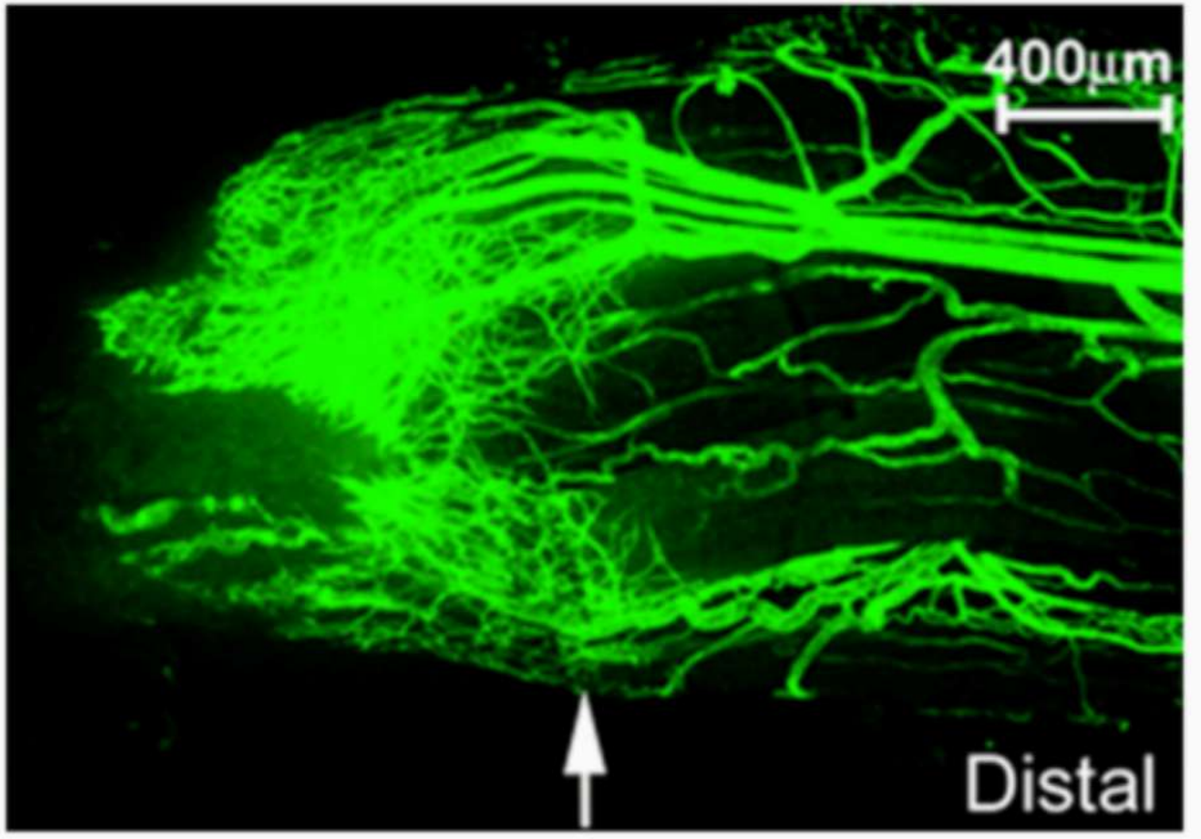
- Hematopoietic stem cells 'home' back to the bone marrow
- Clonal Hematopoietic (CH) cells lose their homing ability
- **Cell movement is not Invasion!**

Tumor cells break through the ECM and move towards blood vessels

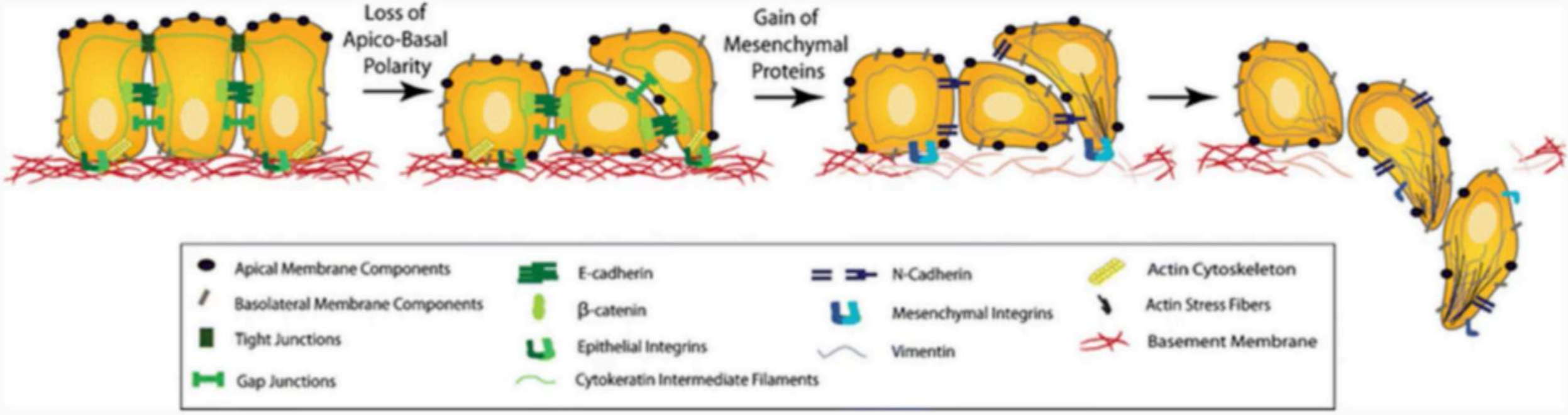
HALLMARK IX: ANGIOGENESIS INDUCTION

- The new blood vessels have a poor morphology and are often leaky

- They are always positioned at a distance from the main vessels to avoid detection.



KEY STEP 1 | INVASION



KEY STEP 2 | INTRAVASATION

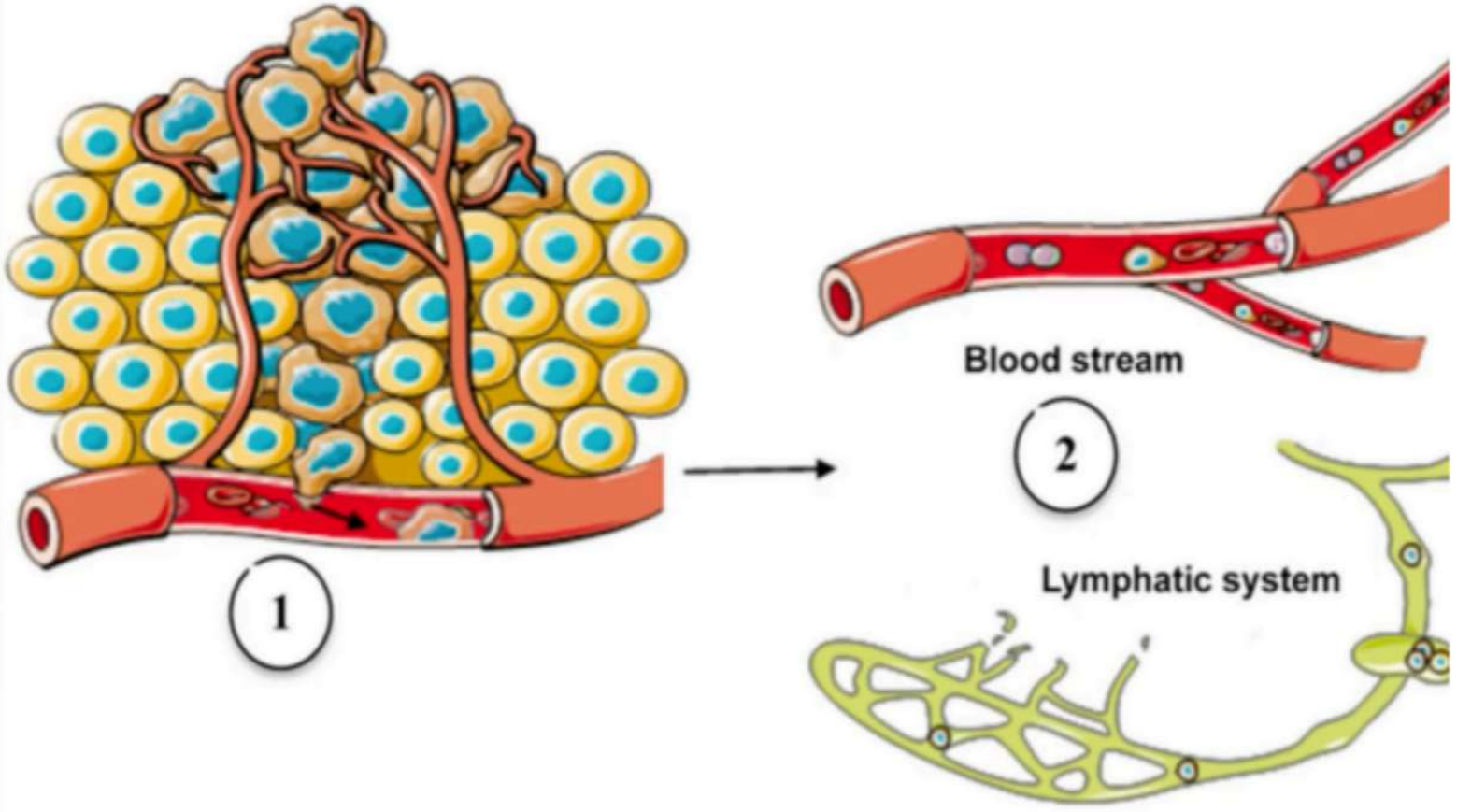
Intravasation is a phenomenon in which a cancer cell enters the blood vessel (#9)

There are two types of intravasations

1. Active Intravasation
2. Passive Intravasation

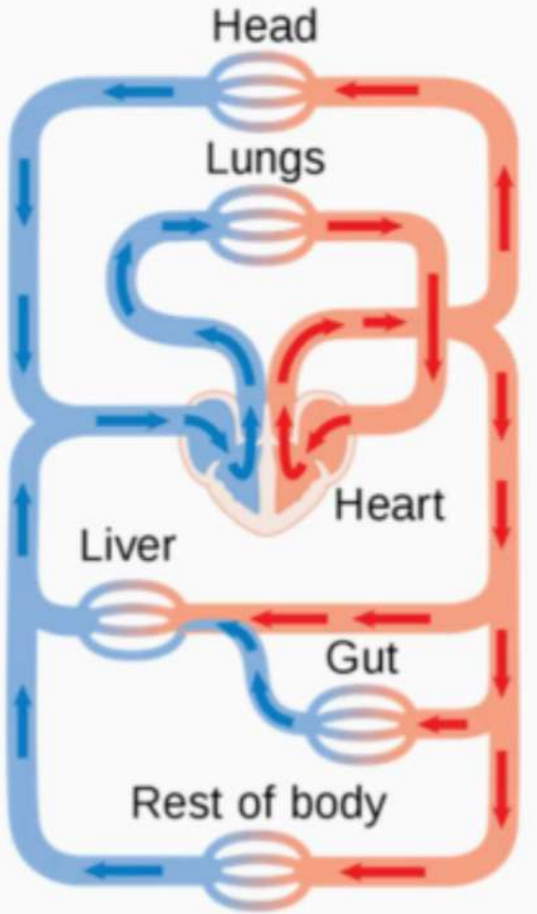
KEY STEP 2 | INTRAVASATION

- Passive Intravasation is the entrance of cancer cells into the blood stream



KEY STEP 3 | SURVIVE SYSTEMIC CIRCULATION

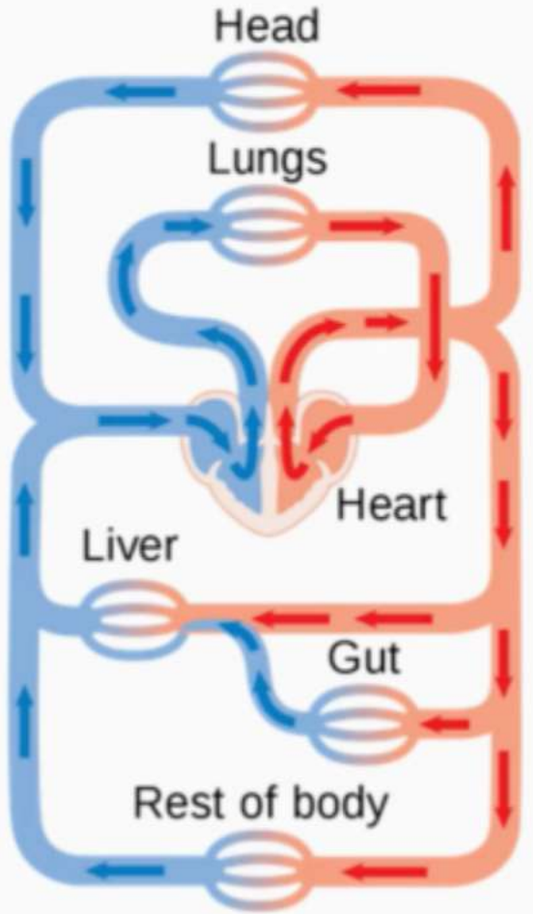
For a cancer to metastasize, it must survive circulation through the bloodstream



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KEY STEP 3 | SURVIVE SYSTEMIC CIRCULATION

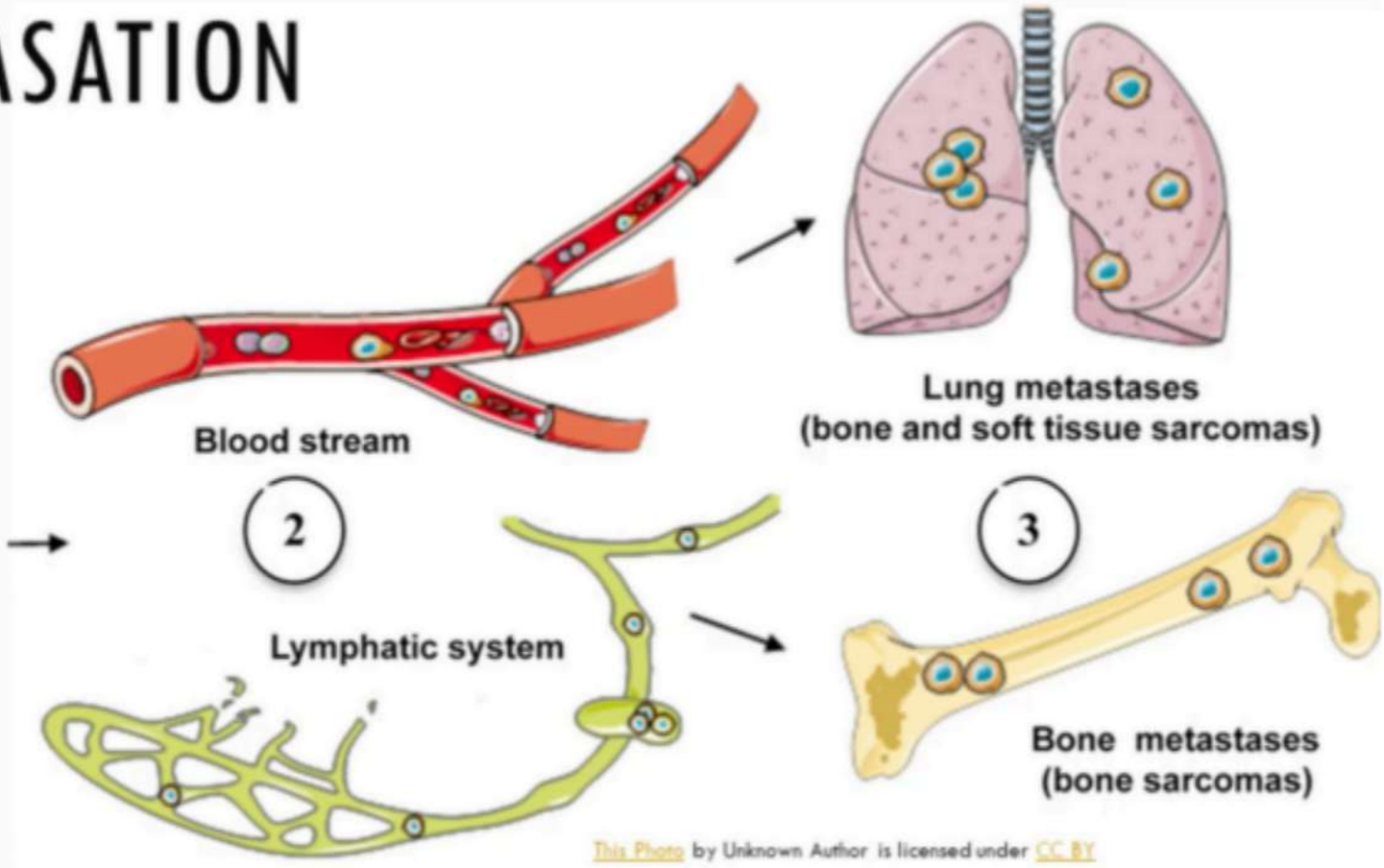
Tumor cells in the blood are called Circulating Tumor Cells (CTCs). If a CTC stops in an organ, it will cause metastasis of that



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KEY STEP 4 | EXTRAVASATION

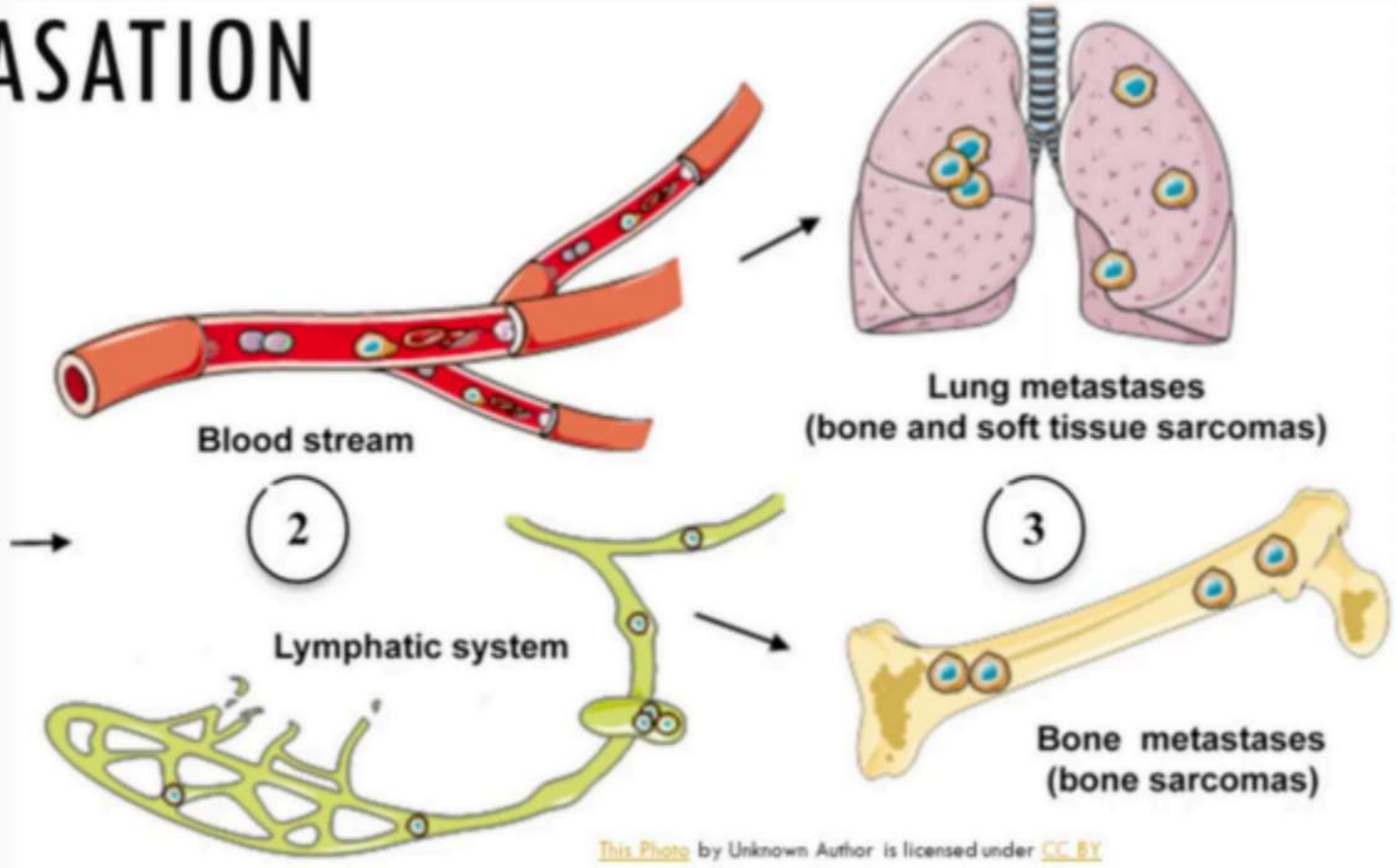
The tumor cells leave the lymphatic system and start setting up secondary tumor sites in a distant organ.



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KEY STEP 4 | EXTRAVASATION

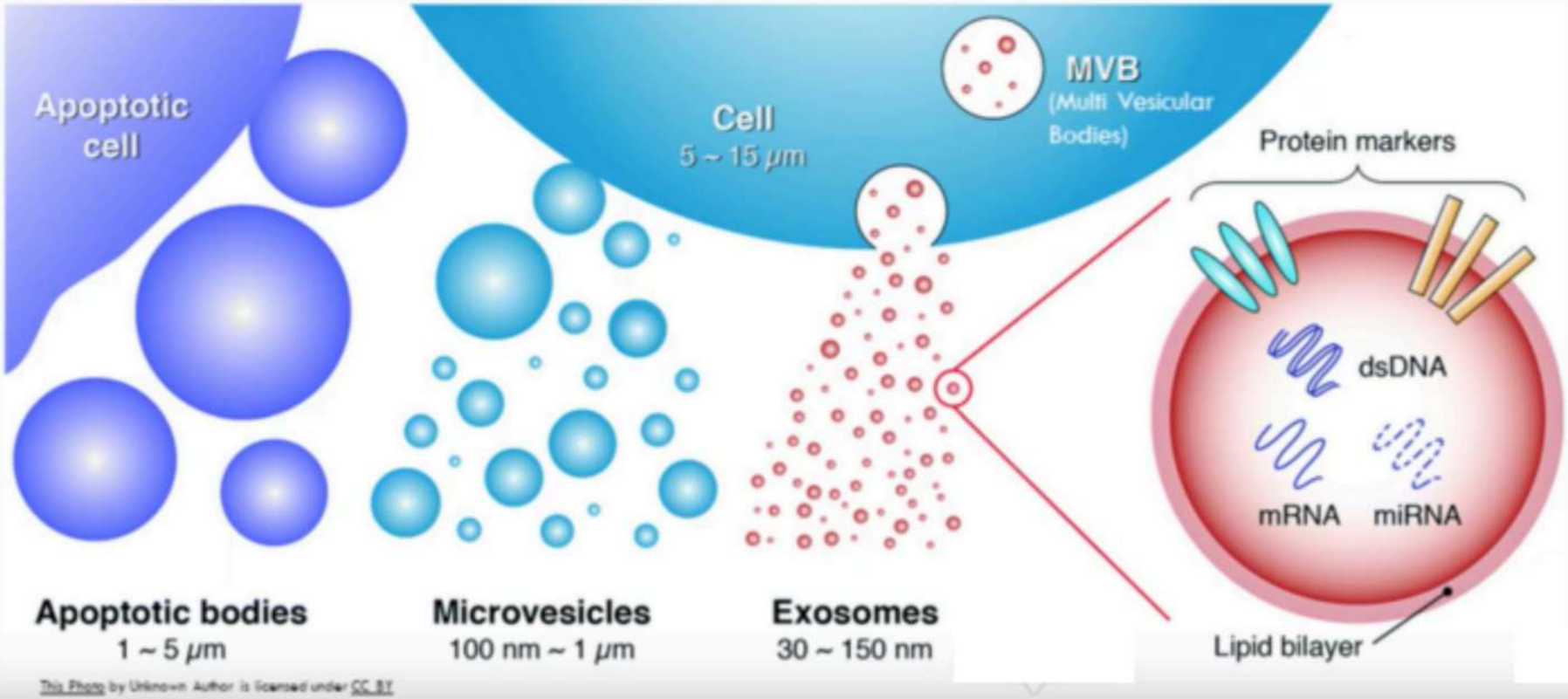
Secondary tumors may not proliferate immediately in the new destination.



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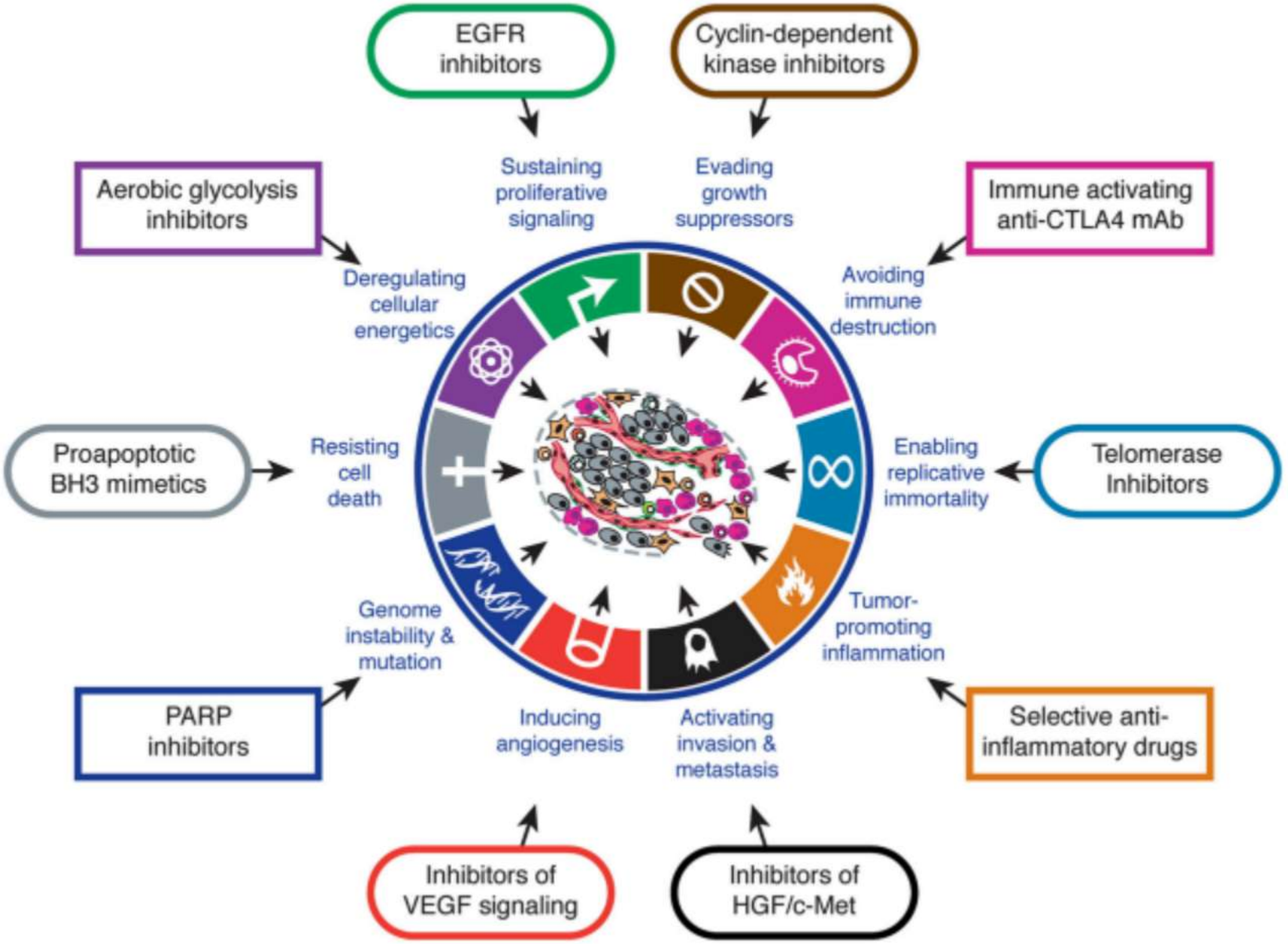
KEY STEP 4 | EXTRAVASATION

Exosomes are EC vesicles that have been recently linked to cancer metastasis. They prepare the groundwork for cancer metastasis sites.



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Therapeutic Targeting of the Hallmarks of Cancer



References

- Hanahan D, Weinberg RA. The hallmarks of cancer. Cell. 2000 Jan 7;100(1):57-70. doi: 10.1016/s0092-8674(00)81683-9. PMID: 10647931.
- Hanahan D, Weinberg RA. Hallmarks of cancer: the next generation. Cell. 2011 Mar 4;144(5):646-74. doi: 10.1016/j.cell.2011.02.013. PMID: 21376230.


THANK YOU

REFERENCES

- Introduction to Cancer Biology, Johns Hopkins School of Medicine
- 30th Annual Short Course on Experimental Models of Human Cancer, The Jackson Laboratory
- Chapter 3: The Hallmarks of Cancer. DeVita, Hellman, and Rosenberg's Cancer: Principles & Practice of Oncology, 11th Edition

(If anyone wants study material regarding cancer, feel free to ask.)



Screenshot


Voice of Dr Peter Fraser
The Babraham Institute
Screenshot

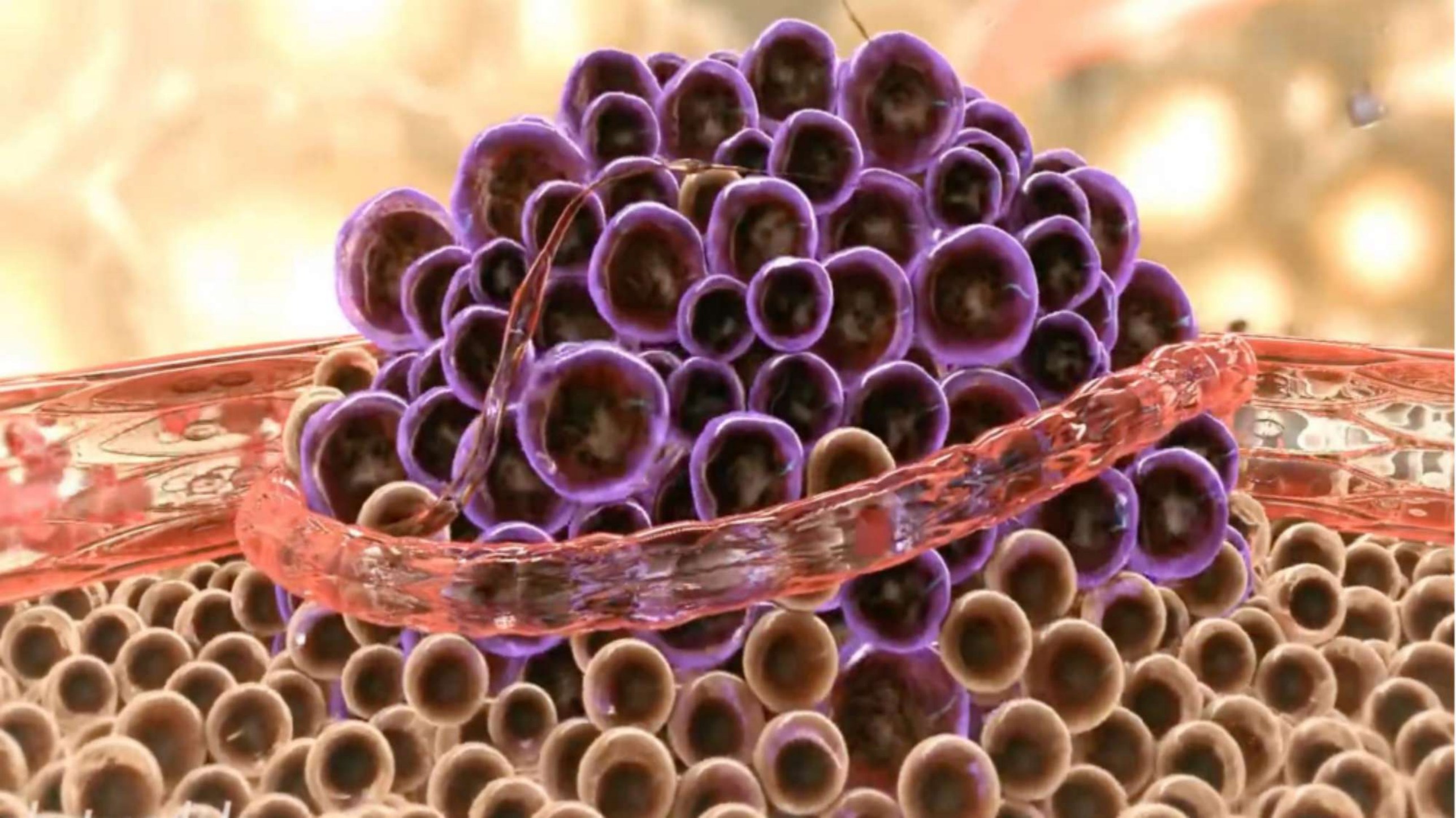
Targeting Oncogenes is Shooting horses

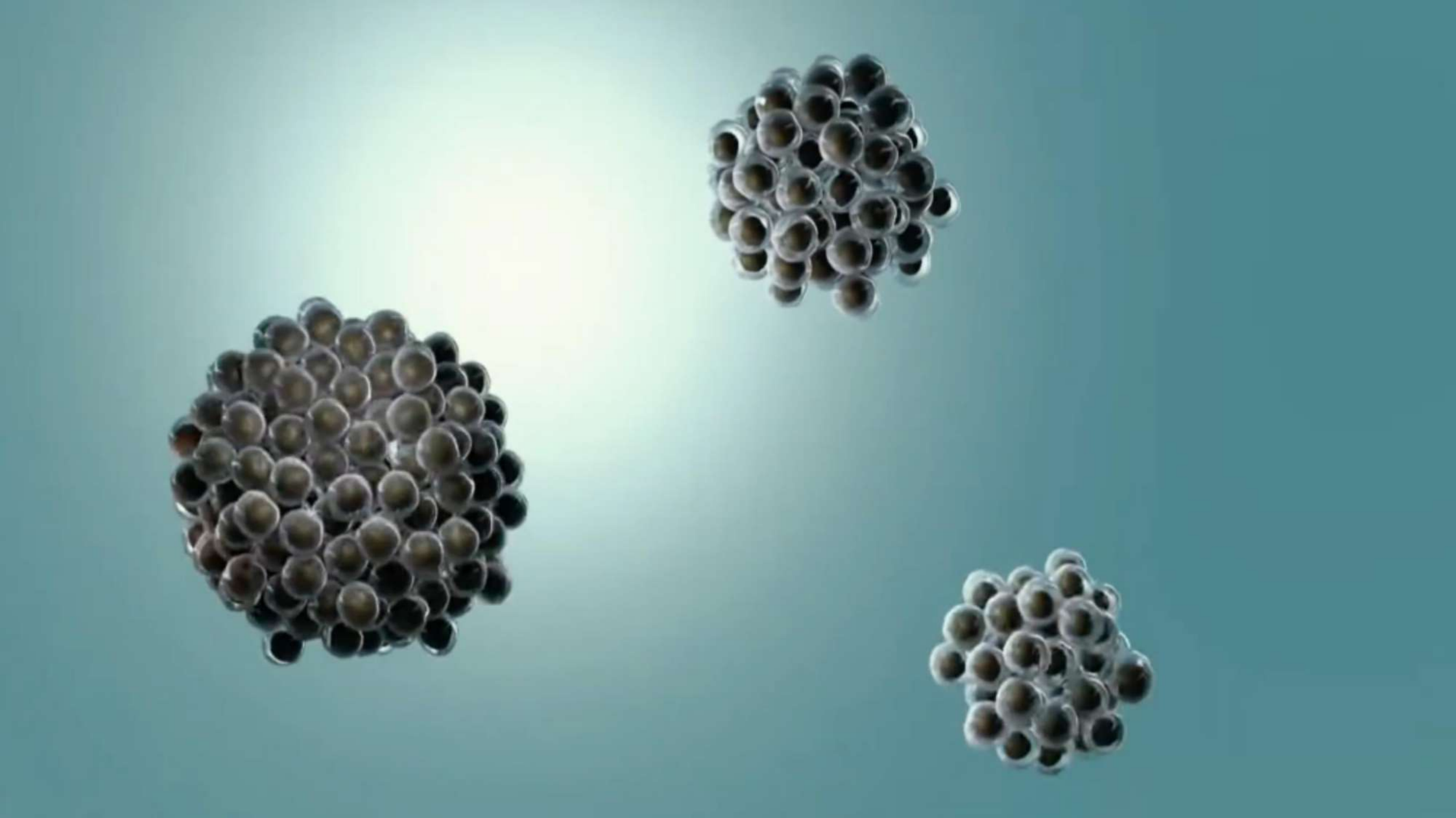


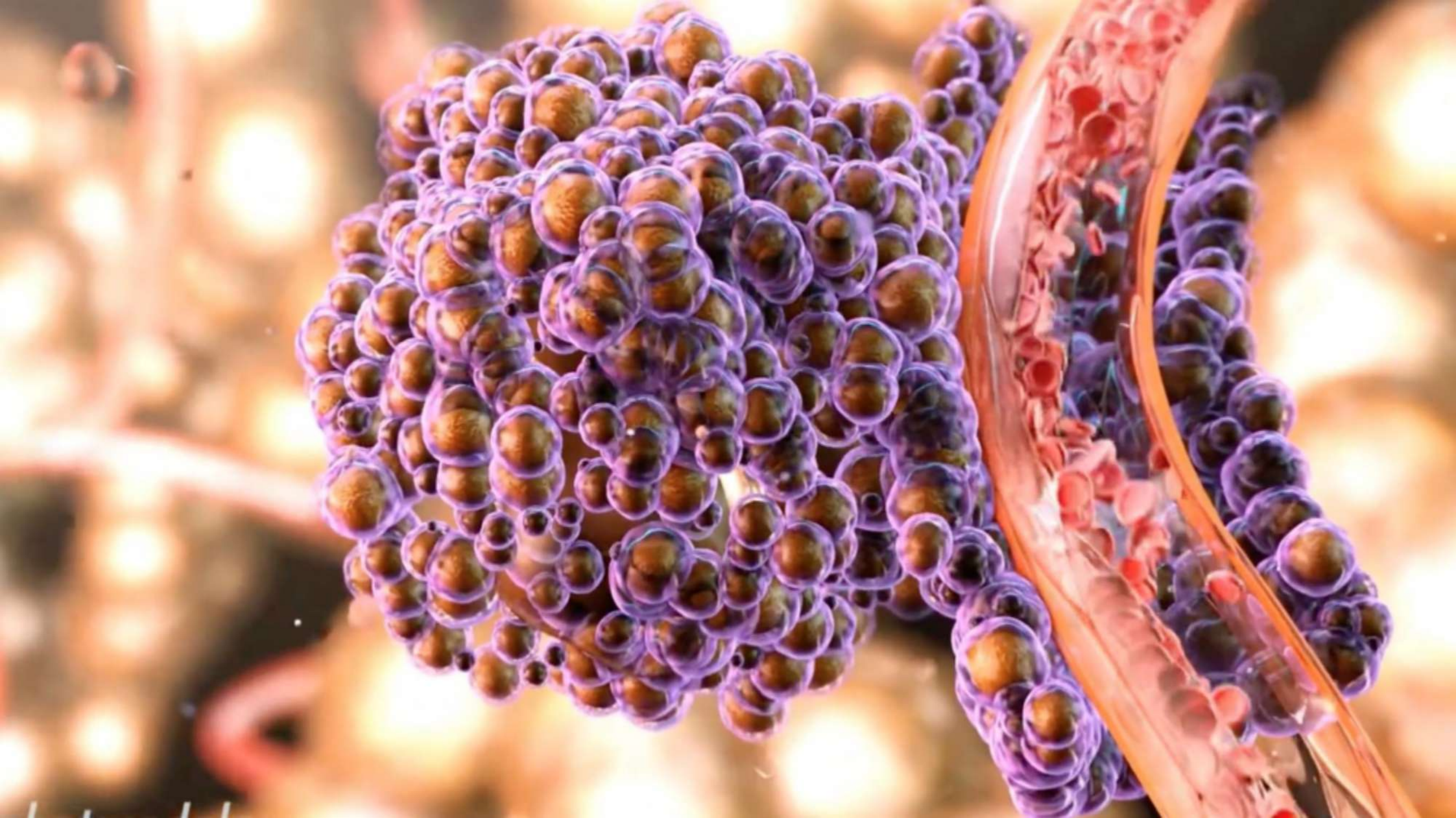
1998 Her-2, Herceptin etc
2001 Bcr-abl, Imatinib etc
2003 EGFR, Gefitinib etc
2003 Proteasome, Bortezomib
2004 VEGF, Bevacizumab
2006 HDAC, Vorinostat
2007 mTOR, Temsirolimus
2010 Provenge
2011 anti-CTLA4
2011 ALK, Crizotinib
2011 B-raf, Vemurafenib
2012 Hedgehog, Vismodegib
2013 Btk, Ibrutinib
2014 PI3k-delta
2014 PARP Olaparib
2014 anti-PD1, Nivolumab
2015 CDK4/6
2016 Bcl-2
2017 Flt3
2017 IDH2/IDH1
2019 PI3-K

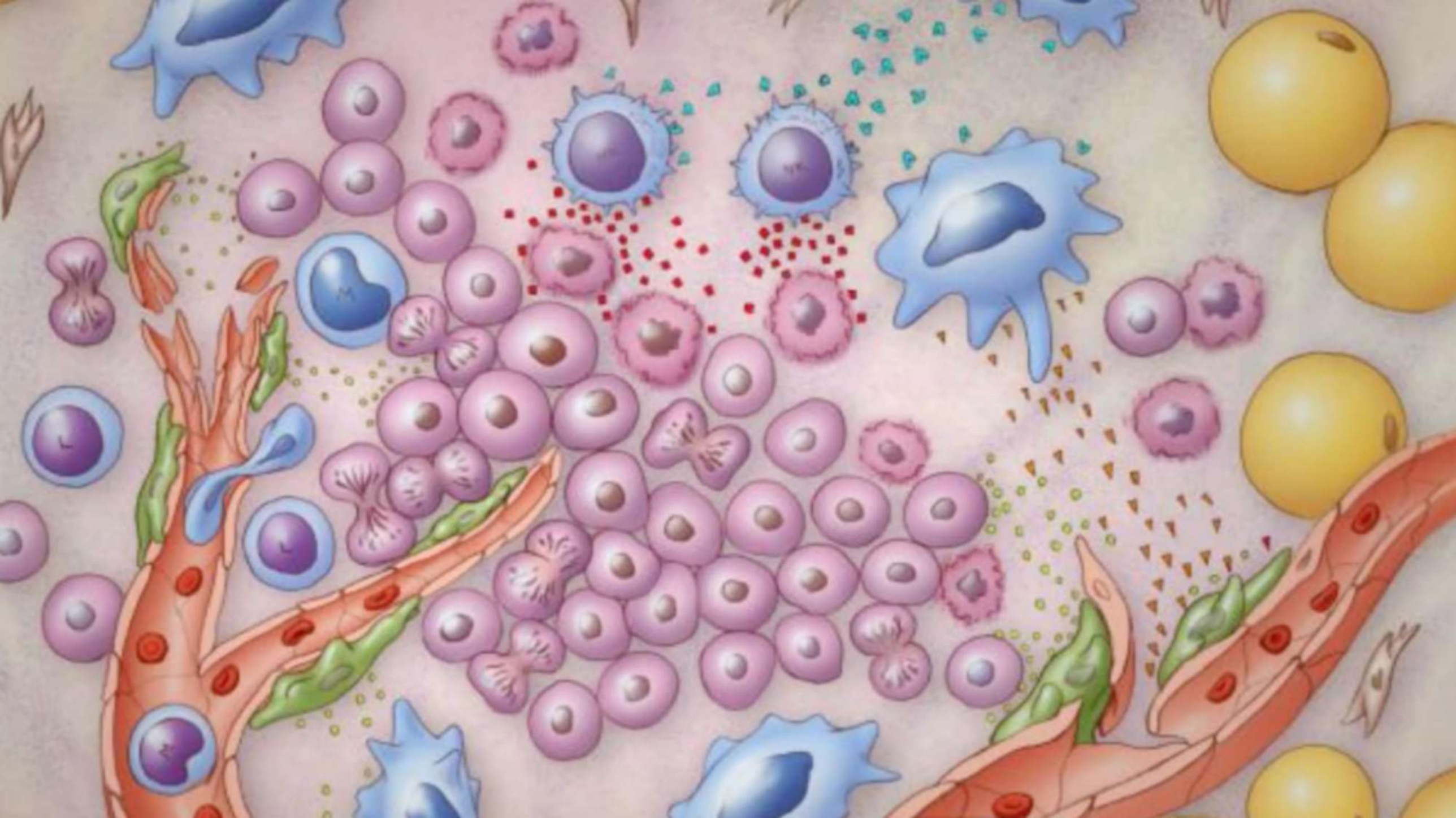
Dr. Mak
University of Hong Kong
Princess Margaret Cancer Center, Toronto
18th August 2021

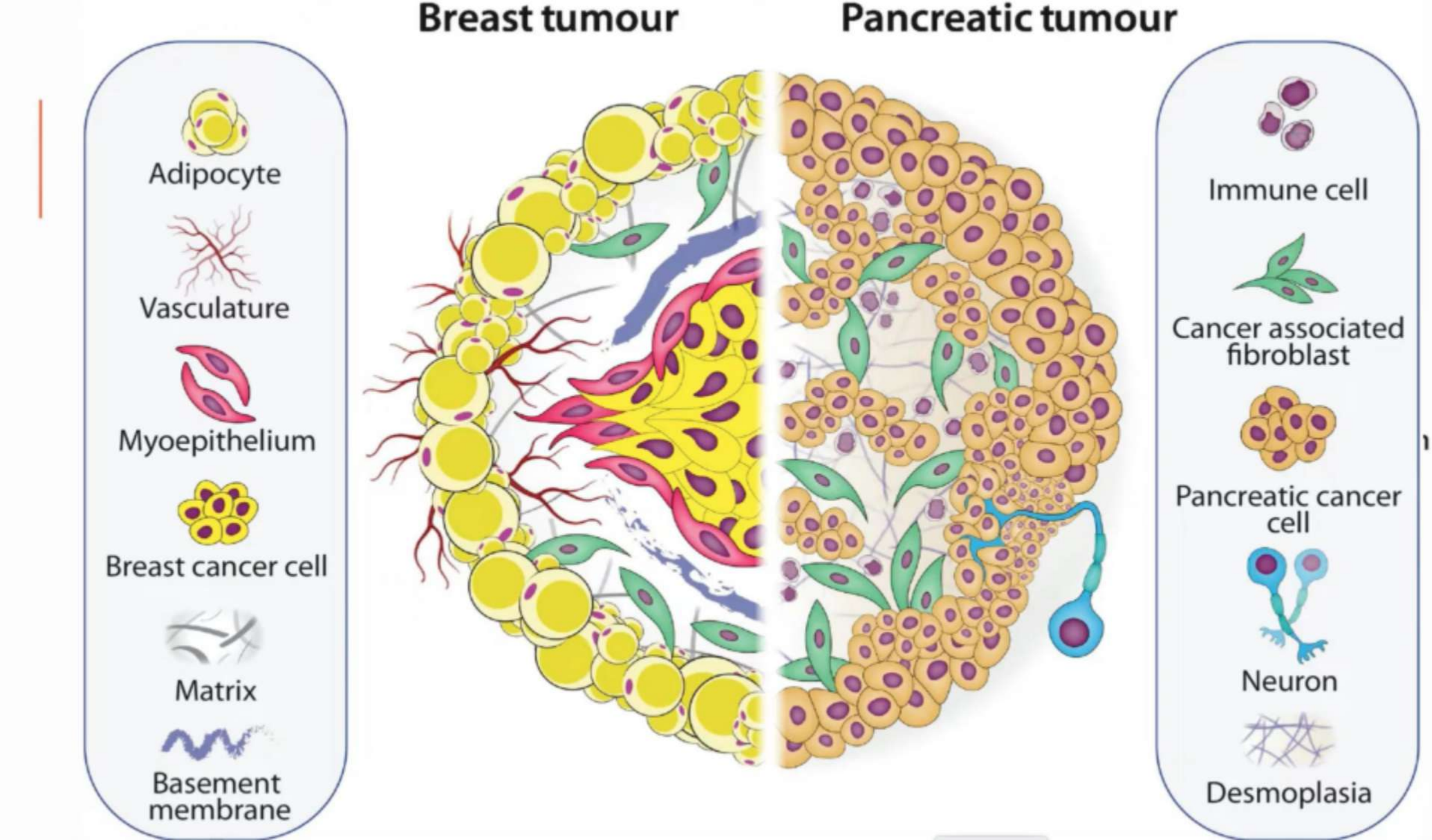
Screenshot



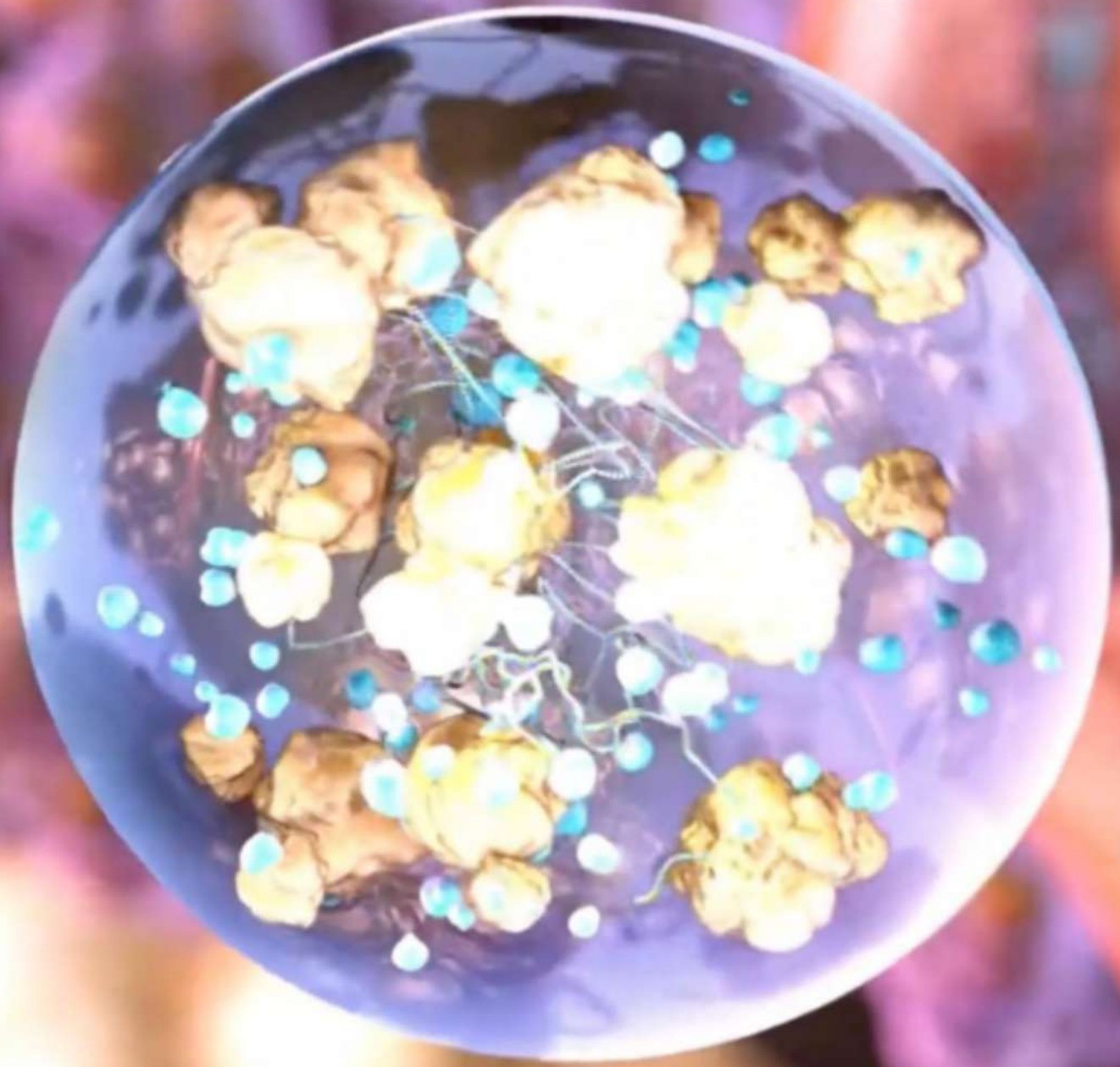







Breast tumour
Pancreatic tumour
Adipocyte
Vasculature
Myoepithelium
Breast cancer cell
Matrix
Basement membrane
Immune cell
Cancer associated fibroblast
Pancreatic cancer cell
Neuron
Desmoplasia

exosome
(50-100 nm)



anobotmodels

TERMINOLOGIES

Fibroblasts : Connective tissues that secrete collagen

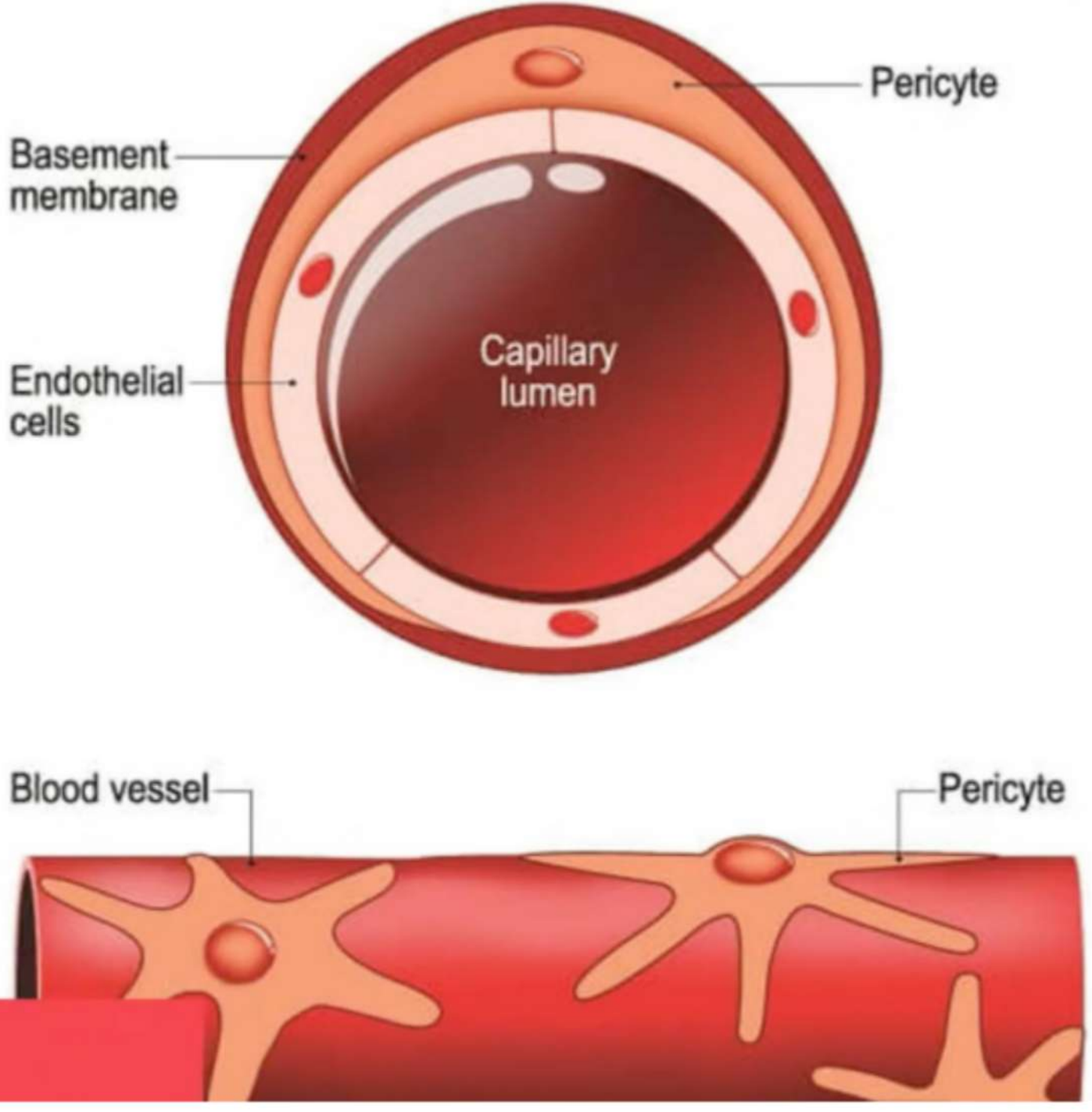
Mesenchyme: The ECM that gives rise to the connective tissues

ECM: Extra cellular matrix – the non-cellular part of the cell

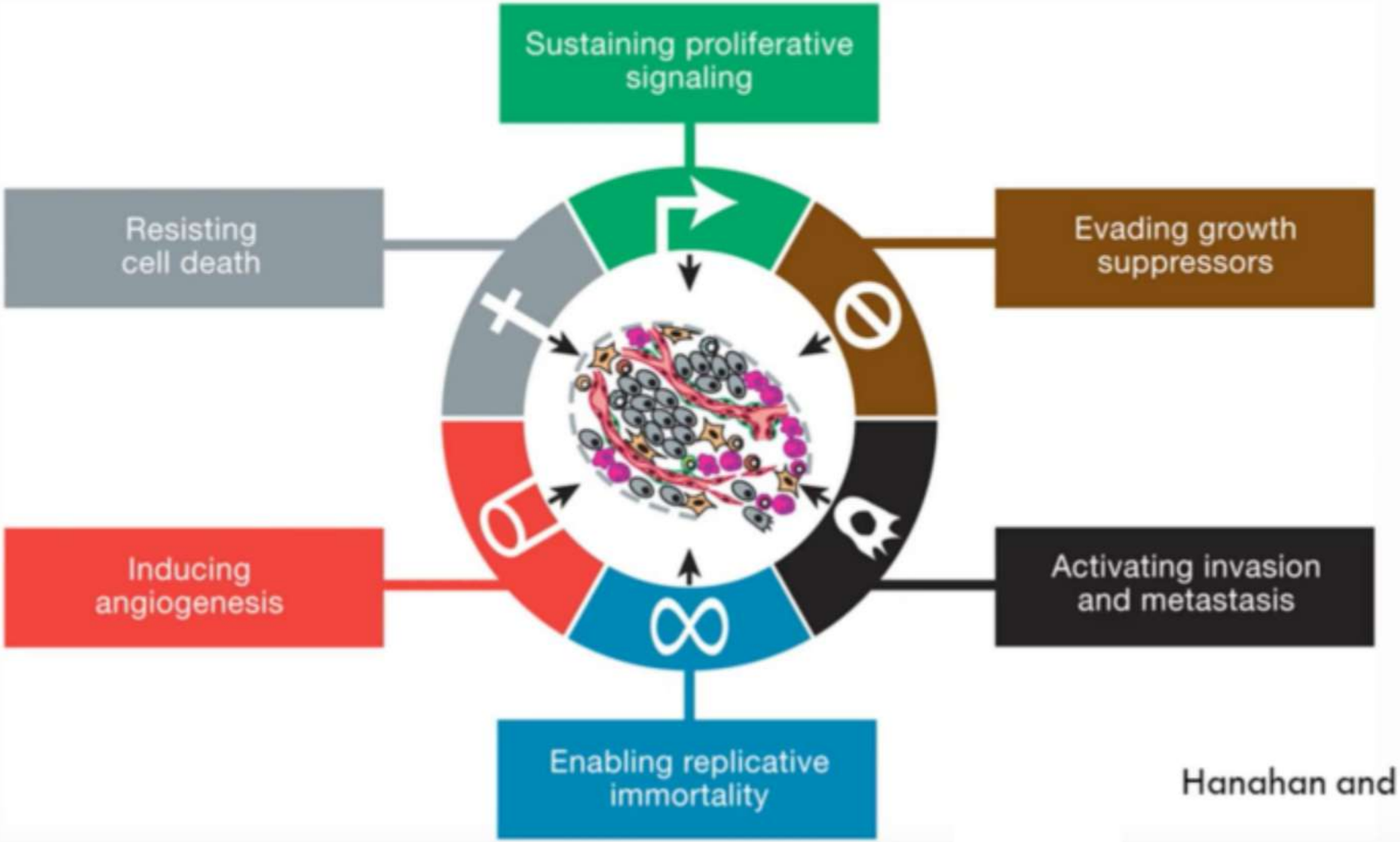
Senescence: The inability to proliferate but remain metabolically active

Pericytes: Cells present along the walls of capillaries at small intervals

Neoplasia: Uncontrolled, abnormal growth of cells/ tissues


Pericyte
Basement membrane
Endothelial cells
Capillary lumen
Blood vessel
Pericyte

THE HALLMARKS OF CANCER 1.0



Hanahan and Weinberg (2000)

Component	Acquired Capability	Example of Mechanism
	Self-sufficiency in growth signals	Activate H-Ras oncogene
	Insensitivity to anti-growth signals	Lose retinoblastoma suppressor
	Evading apoptosis	Produce IGF survival factors
	Limitless replicative potential	Turn on telomerase
	Sustained angiogenesis	Produce VEGF inducer
	Tissue invasion & metastasis	Inactivate E-cadherin